

Tummy Tribe Project

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1 Abstract

Early infancy is a critical period for the gut microbiome development, which can be influenced by age, diet, geography, and medical interventions. In this project, we analyzed longitudinal 16S rRNA gene amplicon sequencing data from a subset of the DIABIMMUNE cohort to characterize age-related changes in the infant gut microbiota between 4, 7, and 10 months of age and to assess the impact of covariates including feeding type, geographical location, and treatment exposure.

Sequencing data were processed in QIIME 2 using DADA2 for denoising, SILVA-based taxonomic classification, and de novo phylogenetic tree construction. Microbiome structure was assessed using alpha and beta diversity metrics, longitudinal mixed-effects models, differential abundance analyses (ANCOM-BC, MaAsLin2, DESeq2), core microbiota analysis, and functional profiling with PICRUSt2.

We observed strong age-associated shifts in microbial composition, with increasing diversity and enrichment of anaerobic genera such as *Blautia*, *Roseburia*, and *Thomasclavelia*. Feeding type and geographic location were associated with distinct community structures and core taxa, whereas treatment exposure showed more limited effects.

Overall, age was the main factor associated with gut microbiome development in infants, whereas diet and geographic location influenced the microbiome to a lesser extent.

2 Introduction

The gut microbiota of infants develops and changes a lot in the first year of life mostly as they transition from milk to solid foods [1]. The changes are driven by both environmental and genetic factors such as age, diet, environmental exposure, and medical interventions. The infant gut microbiota in early life goes from being dominated by *Bifidobacterium* and other milk-adapted taxa to a more diverse community enriched in members of the phyla *Firmicutes* and *Bacteroidota*. [2]. In this project, we analyze a novel cohort of infant fecal samples using 16S rRNA gene sequencing data processed in QIIME 2, together with detailed sample metadata. The data used in this project is a subset of the DIABIMMUNE cohort. The DIABIMMUNE study recruited about 1000 infants from Espoo (Finland), Petrozavodsk (Russia), and Tartu (Estonia). Extensive metadata was collected along with monthly stool samples from birth until the children were 3 years old [3]. In this analysis, we focus on samples collected at 4, 7, and 10 months of age, capturing key stages of early gut microbiota maturation. Our aim is to characterize age-related changes in the infant gut microbiota and to assess the influence of covariates, including diet, geographical origin, and treatment exposure. Specifically, we investigate differences in microbial composition across age groups, identify potential core microbiota associated with breastfed and formula-fed infants, and evaluate whether treatment exposure affects gut microbiome structure and whether such effects depend on other covariates.

3 Methods

3.1 Data and General Workflow

We were given sequencing data consisting of paired-end, demultiplexed reads with Phred quality scores from the V4 region of the 16S rRNA gene. The sequences were generated using an Illumina MiSeq platform. We had 357 samples from 190 children. The metadata table contains the sample id, host id and information about the age in months, geographical location, delivery mode, sex, diet weaning, diet milk and treatment exposure.

Figure 1 shows the general workflow of our analysis.

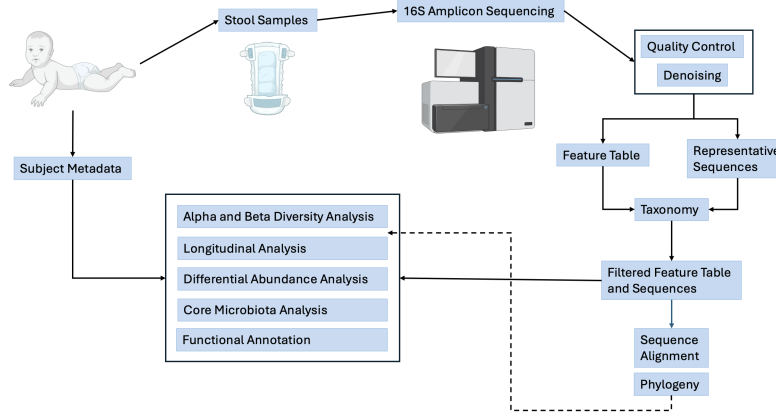


Figure 1: Workflow

3.2 Quality Control and Denoising

Sequence quality was inspected for trimming and filtering decisions. Forward reads generally showed acceptable quality, whereas reverse reads displayed a strong decline in quality after approximately 150 nucleotides, indicating that truncation was required (Supplementary Figure 19).

Denoising was performed using DADA2 [4]. After evaluating several truncation lengths, we chose 135 base pairs for forward and reverse reads. Shorter truncation lengths resulted in insufficient overlap for successful paired-end merging for most reads, while longer truncation lengths retained more low-quality bases during filtering, leading to increased read loss during denoising.

All other DADA2 parameters were kept at default settings. Reads were filtered, denoised, merged, and screened for chimeras, resulting in a feature table of amplicon sequence variants (ASVs) and representative sequences. The final dataset contained 3,873 ASVs and approximately 16 million reads.

3.3 Taxonomy

Taxonomic classification was compared using three Naive Bayes classifiers based on the SILVA reference database. These included a region-specific classifier trained on V4-trimmed reference sequences, a classifier trained on full-length SILVA sequences, and a pretrained SILVA classifier obtained from the QIIME 2 data resources and weighted for human stool samples [5–9].

Custom classifiers were trained in QIIME 2, with computationally intensive steps performed on the Euler high-performance computing cluster. Classifier performance was compared using the QIIME 2 `evaluate-fit-classifier` method.

The region-specific classifier assigned the highest proportion of sequences to deeper taxonomic levels and showed the highest taxonomic entropy, and was therefore used for all downstream taxonomic analyses.

3.4 Phylogeny

We aligned 16S rRNA genes across divergent taxa to build a de novo tree [10]. We used alignment masking, which removes ambiguously aligned regions. To then construct the phylogenetic tree we

used FastTree from the QIIME 2 phylogeny plugin [11]. To assert the robustness of our tree we used bootstrapping.

3.5 Diversity Analysis

Alpha and beta diversity metrics were calculated in QIIME 2, including phylogenetic measures such as UniFrac, using the constructed phylogenetic tree [12]. A rarefaction depth of 14,000 reads per sample was selected to balance sample retention and adequate representation of community structure.

Beta diversity was statistically tested using PERMANOVA (adonis2) [13]. To account for repeated measurements in the dataset, infant ID was included as a strata variable. Variables that do not vary within infants (delivery mode, geographic location, treatment exposure) were tested by running PERMANOVA separately for each age group using one sample per infant.

Alpha diversity was analyzed using linear mixed-effects models in the longitudinal analysis. Kruskal-Wallis test results were not interpreted because this test cannot account for repeated measurements within infants.

3.6 Longitudinal Analysis

Longitudinal analyses were conducted to assess temporal changes in the infant gut microbiome. First differences in alpha diversity and first distances in beta diversity were calculated between consecutive sampling time points. Volatility plots were used to visualize diversity trajectories over time.

Alpha diversity was analyzed using linear mixed-effects (LME) models with infant ID included as a random effect. Feature volatility analysis was used to identify taxa associated with age, and selected taxa were subsequently analyzed using LME models.

3.7 Differential Abundance Analysis

We performed differential abundance analysis using ANCOM-BC from QIIME 2 [14], MaAsLin2 in R [15, 16], and DESeq2 in R [17, 18]. Because each method captures different aspects of the data, we generated a consensus set of taxa that are consistently detected across all methods. We filtered the data to only include ASVs that have a total abundance of at least 25 counts across all samples and are present in at least 4 samples. We used Benjamini-Hochberg (BH) for p-value adjustment across all methods. A taxon was defined as differentially abundant if the log-fold-change was greater than 1 and the adjusted p-value was smaller than 0.05. For MaAsLin2 we used 'TSS' (Total Sum Scaling) for the transformation and log-normalized our data. For DESeq2 we used DESeq's built-in normalization (type="poscounts" for sparse microbiome data) and a negative binomial GLM with Wald test.

3.8 Core Microbiota Analysis

Core microbiota analyses were performed to compare core genera across delivery mode, feeding pattern, treatment exposure, and geographic location. Core taxa were identified using the QIIME 2 **core-features** method [19]. Analyses were conducted separately for each age group, using a single sample per child to avoid bias from repeated measurements.

For each age group and metadata variable, samples were stratified into groups and genera were defined as core based on their prevalence within each group. Core taxa were identified starting at a minimum prevalence threshold of 50%, with thresholds increased in 5% increments to capture the effect of different core definitions.

Genera were then classified according to their core status within each group (core in one group only, core in both groups, or not core in either group), and these patterns were visualized using heatmaps.

3.9 Functional Profiling

Functional profiling was performed using the PICRUSt2 pipeline [20]. Amplicon sequences were placed into a reference phylogeny using EPA-ng, followed by hidden state prediction via maximum parsimony to infer metagenome and pathway-level abundances. The representativeness of

the functional predictions was assessed using Nearest Sequenced Taxon Index (NSTI) scores. Functional beta diversity analysis was conducted on MetaCyc pathway abundances using QIIME 2 core beta-diversity metrics and the same statistical tests as the taxonomic beta diversity analysis (see section 3.5). Finally, differentially abundant pathways across metadata variables were identified using ANCOM-BC.

3.10 Use of Artificial Intelligence Tools

AI tools were only used to support code development, debugging, and language editing in the Jupyter notebooks. We conducted and validated all data analyses, results, and scientific interpretations.

4 Results

4.1 How do the microbiome samples differ between different ages?

To assess how the gut microbiome changes during the first year of life, we first examined the distribution of samples across age groups. Sample numbers were uneven between ages, and many children contributed only a single sample (Supplementary Figure 14).

We then performed beta diversity analyses, which showed a significant association between age and beta diversity across all distance metrics (PERMANOVA, $p = 0.001$) as shown in Figure 2. The explained variation (R^2) is highest for phylogenetic metrics, especially unweighted UniFrac ($R^2 = 0.0688$) and weighted UniFrac ($R^2 = 0.0471$). In contrast, non-phylogenetic metrics (Jaccard and Bray–Curtis) show lower R^2 values (≈ 0.02), reflecting smaller but still significant changes in community membership and abundance. Overall, these results suggest that microbiome development with age involves changes in both, which taxa are present and how they are related, with phylogenetic changes playing a bigger role.

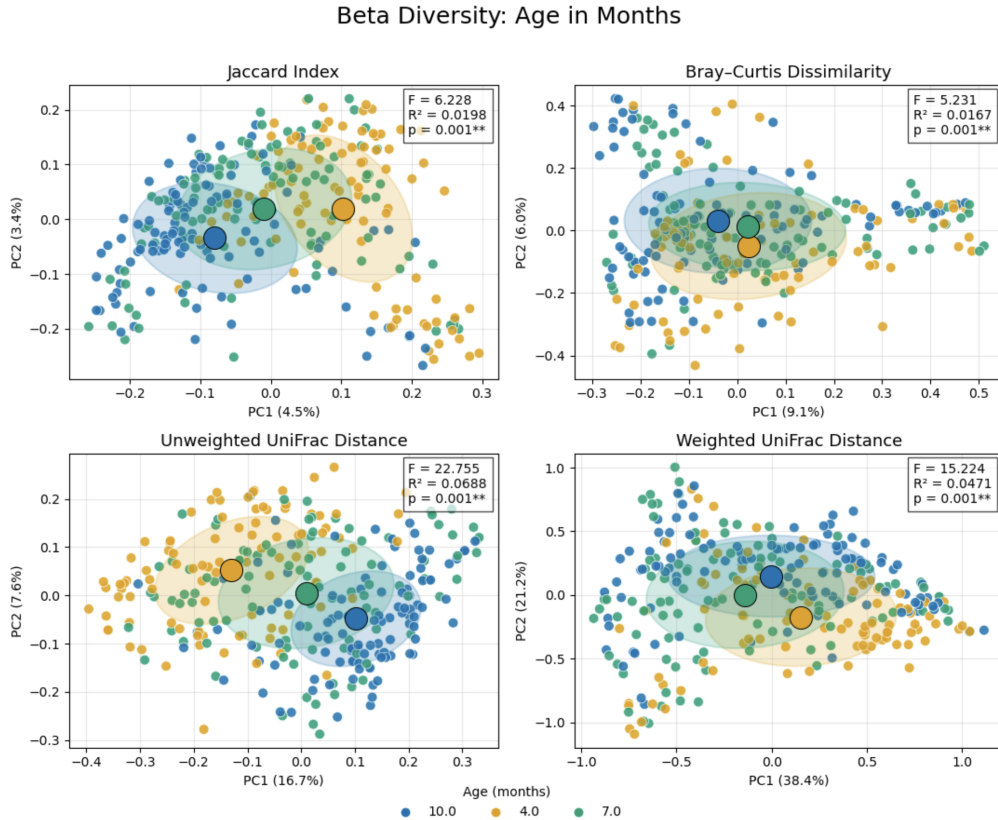


Figure 2: **PCoA plot for beta diversity patterns stratified by age.** Each plot shows PCoA for one beta-diversity metric. Each dot represents one sample, samples are coloured by age.

All results are reported at the genus level, where SILVA classification is most reliable.

Longitudinal analysis showed that alpha diversity increased with age across individuals (Supplementary Table 9). Feature volatility analysis identified *Blautia* and *Thomasclavelia* as the taxa with the strongest age-related patterns. Formal longitudinal modeling provided evidence for an age-related increase in *Blautia*, but did not reveal clear differences in taxon trajectories across metadata groups (Supplementary Table 12).

Differential abundance analysis identified 11 taxa that differed between the 4 and 10 month age groups (Figure 3, Table 1). Several genera, including *Thomasclavelia*, *Roseburia*, and *Blautia*, increased with age, consistent with the longitudinal results. Age-related increases in *Roseburia* and *Blautia* have also been reported previously [21]. In contrast, *Staphylococcus* and *Lactacaseibacillus* were the only taxa that decreased with age.

Functional profiling showed enrichment of the CODH-PWY pathway at 10 months compared with 4 months (Figure 4), corresponding to the reductive acetyl-CoA pathway, a key anaerobic carbon fixation pathway. Across analyses, the consistent increase of anaerobic taxa such as *Blautia*, *Thomasclavelia*, and *Roseburia*, together with the enrichment of CODH-PWY, suggests a shift toward a more mature, anaerobic gut ecosystem with age.

Table 1: **Differentially expressed genera in age.** Down means it's depleted in age group 10 months.

Genus	DESeq2 lfc	MaAsLin2 lfc	ANCOM-BC lfc	direction
<i>Thomasclavelia</i>	1.22	3.76	1.85	up
<i>Faecalibacterium</i>	2.95	2.98	2.49	up
<i>Flavonifractor</i>	1.94	2.54	1.10	up
<i>Roseburia</i>	3.00	2.75	1.49	up
<i>Lachnospira</i>	2.62	2.89	1.73	up
<i>Blautia</i>	2.50	5.06	3.29	up
<i>Anaerostipes</i>	1.99	3.30	2.11	up
<i>Agathobacter</i>	2.97	2.18	1.40	up
<i>Staphylococcus</i>	-1.83	-1.21	-1.70	down
<i>Lactacaseibacillus</i>	-1.38	-1.60	-2.22	down
<i>Granulicatella</i>	1.26	0.89	-0.38	up

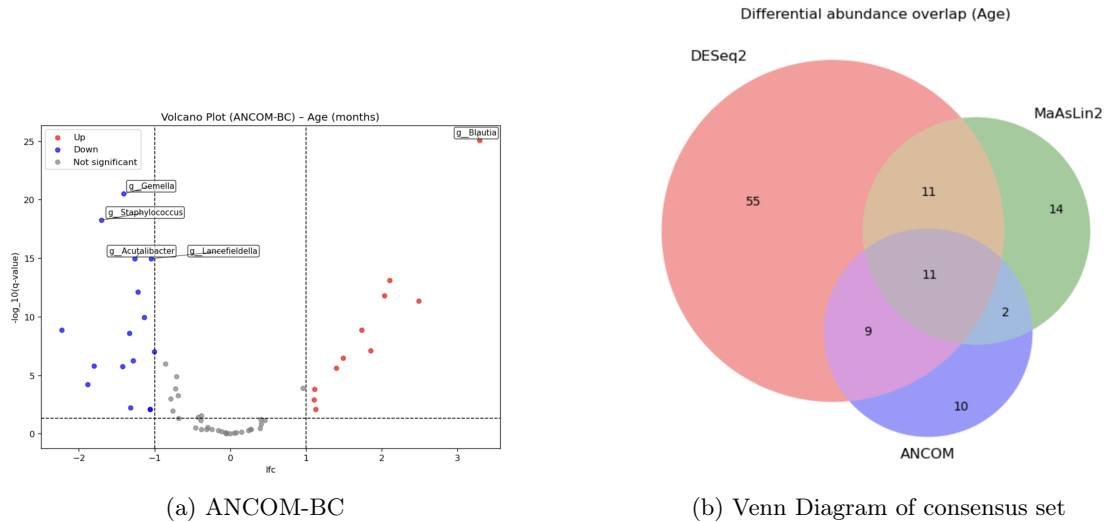


Figure 3: **Differential Abundance Analysis between infants aged 4 and 10 months.** The Volcano Plot from ANCOM-BC shows genus-level differential abundance. The Venn Diagram summarizes overlap between methods, highlighting taxa consistently identified as differentially abundant across approaches. 11 differentially expressed taxa were found by all three methods. Volcano Plots from MaAsLin2 and DESeq2 can be found in the supplementary figures.

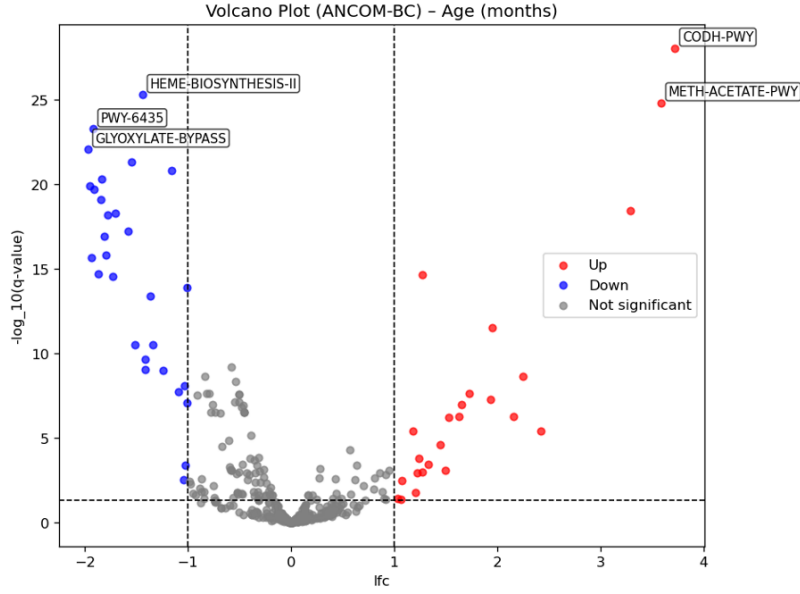


Figure 4: **Volcano plot of differentially abundant pathways associated with age.** Functional differential abundance analysis comparing infants aged 4 and 10 months using ANCOM-BC.

4.2 What differences do you see in terms of geolocation?

First, we assessed the independence of metadata variables. Chi-square tests indicated that geographic location was not independent of milk diet, treatment exposure, or age, and showed a weaker association with weaning diet, but no association with delivery mode or sex (Supplementary Table 5).

To evaluate differences in infant gut microbiome composition across geographic locations, we performed alpha and beta diversity analyses as well as differential abundance testing. Because geographic location is time-invariant, beta diversity was assessed using age-stratified PERMANOVA (Figure 5). Geographic location showed consistent associations with beta diversity across ages, with the strongest effects observed for weighted UniFrac, suggesting differences in the relative abundance of phylogenetically related taxa across locations.

In the differential abundance analysis we found 5 genera that were depleted in Finnish infants compared to other infants and 3 that were enriched (Figure 6, Table 2). This result has to be handled with some care because we could not correct for repeated measurements between infants in DeSeq2. *Bifidobacterium* was found to be more abundant in Russian children than in Finnish children by Vatanen et al. (2016) [3], which matches our results.

Table 2: **Differentially expressed taxa in geographical location.** Down means it's depleted in Russian infants.

Genus	DESeq2 lfc	MaAsLin2 lfc	ANCOM-BC lfc	direction
<i>Schaalia</i>	2.82	2.78	0.84	up
<i>Bifidobacterium</i>	2.79	2.98	1.70	up
<i>Eggerthella</i>	4.38	2.18	1.04	up
<i>Enterococcus</i>	5.99	4.46	3.29	up
<i>Clostridium</i>	-2.44	2.65	-2.38	down
<i>Campylobacter</i>	-3.42	-1.36	-1.21	down
<i>Sutterella</i>	-4.41	-2.19	-0.80	down
<i>Pseudomonas</i>	5.90	1.94	1.030	up

Beta Diversity – Geographic Location

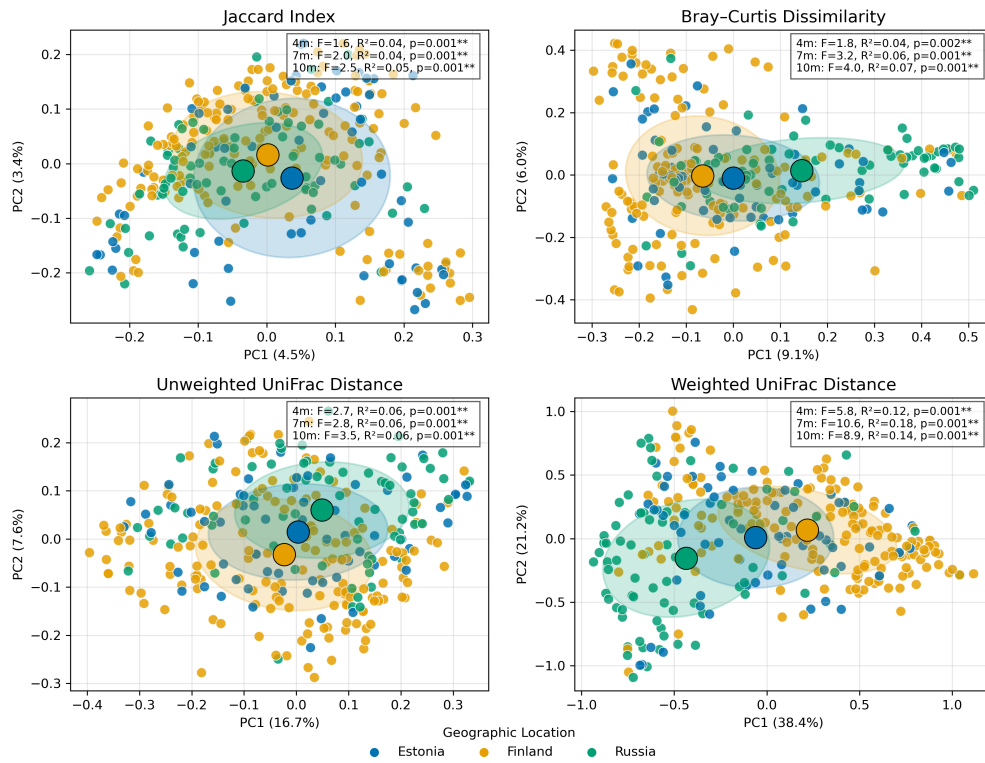


Figure 5: **PCoA plot for beta diversity patterns stratified by geographical location.** Each plot shows PCoA for one bea-diversity metric. Each dot represents one sample, samples are coloured by age.

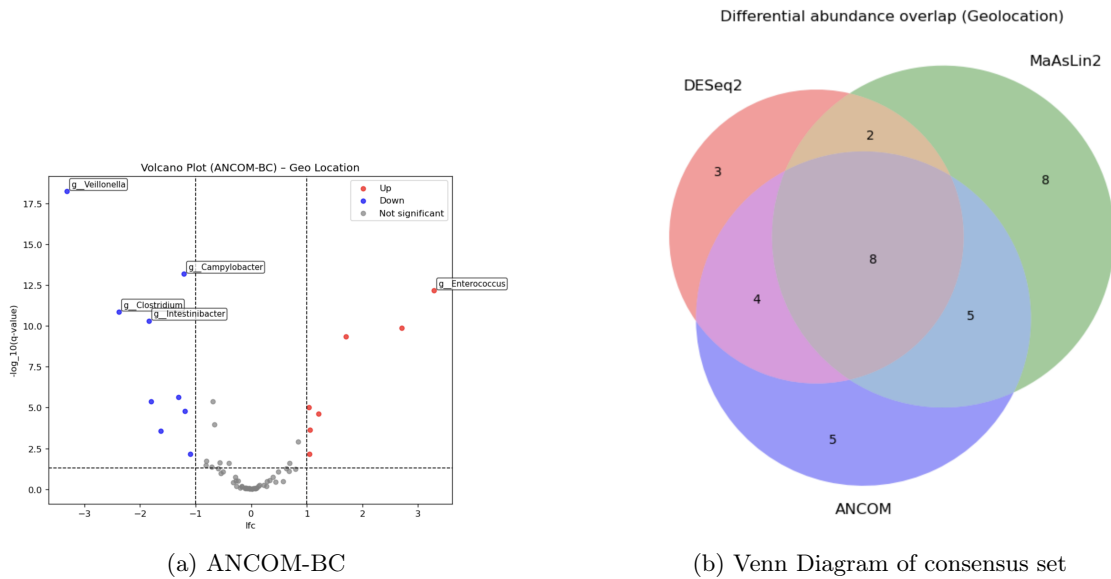


Figure 6: **Differential Abundance Analysis between Finish and Russian infants.** The Volcano Plot from ANCOM-BC shows genus-level differential abundance. The Venn Diagram summarizes overlap between methods, highlighting taxa consistently identified as differentially abundant across approaches. 8 differentially expressed taxa were found by all three methods. Volcano Plots from MaAsLin2 and DESeq2 can be found in the supplementary figures.

4.3 Some of your samples stem from infants that were dominantly fed with breast-milk (diet_milk="bd") and some were fed formula (diet_milk = "fd"). What would be the “core microbiota” of these two groups? How do they differ?

Before comparing breast-fed (bd) and formula-fed (fd) infants, we examined the metadata to assess sample size, balance, and potential covariates. Although three diet categories are present, this analysis focuses on breast-fed versus formula-fed infants. Diet information was missing for a substantial number of samples (87/357), and only 36 samples originated from breast-fed infants, all collected at 4 months of age (Figure 7a, Figure 7b). Feeding type was unevenly distributed across age and geographic location (Supplementary Table 6), indicating potential confounding effects in downstream analyses.

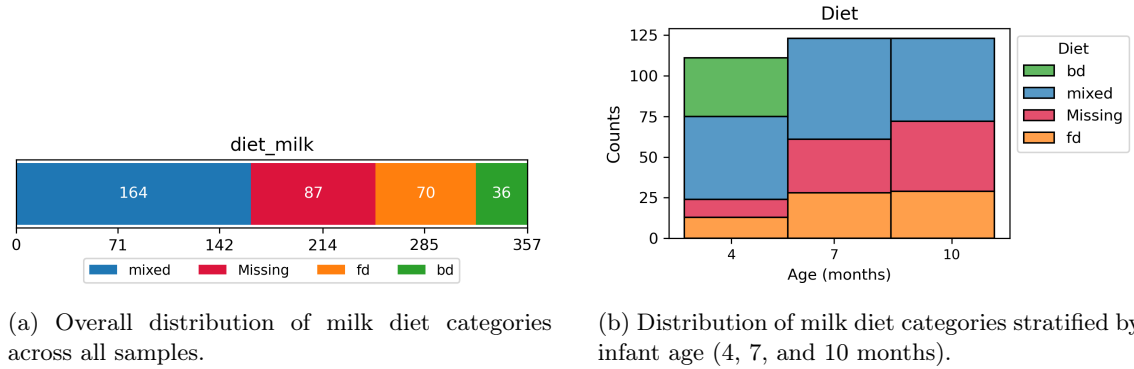


Figure 7: **Overview of infant milk diet metadata.** (a) Total counts of samples per milk diet category, including breast-fed (bd), formula-fed (fd), mixed feeding, and missing information. (b) Age-stratified distribution of milk diet categories, illustrating changes in feeding patterns across early infancy.

To find the core microbiota, we compared breast-fed (bd) and formula-fed (fd) infants at 4 months of age (Figure 8a). Core size decreased with increasing prevalence thresholds in both groups (Figure 8b), suggesting comparable overall core stability. Several genera, including *Bifidobacterium*, *Bacteroides*, *Streptococcus*, and *Veillonella*, were shared between bd and fd infants across most thresholds. This is consistent with previous reports of these genera as early-life core taxa [22, 23].

At the same time, we saw some diet-specific differences. *Haemophilus* was shared between bd and fd at low thresholds, but at stricter thresholds it stayed core only in bd, meaning it was more consistently present in breast-fed infants. *Haemophilus* has been reported as a common genus in breast milk [22], but this pattern could also be influenced by covariates such as geographic location. In contrast, *Escherichia-Shigella* was fd-specific across all thresholds, which matches earlier studies reporting higher *Enterobacteriaceae* in formula-fed infants [23].

To assess overall community differences between breast-fed and formula-fed infants, we analyzed beta diversity (Figure 9). All distance metrics showed significant differences (PERMANOVA, $p = 0.001$), with strongest effects for phylogenetic measures, suggesting that milk type influences both microbial composition and phylogenetic structure.

To determine which taxa contribute to the diet-related differences observed above, we performed a differential abundance analysis (Table 3). Two genera were consistently identified across methods: *Blautia* and *Anaerostipes*, both belonging to the family *Lachnospiraceae*. Both genera were enriched in formula-fed infants, suggesting a diet-associated increase of *Lachnospiraceae* in this group. This finding is in line with earlier reports of higher *Lachnospiraceae* abundance in formula-fed infants [24].

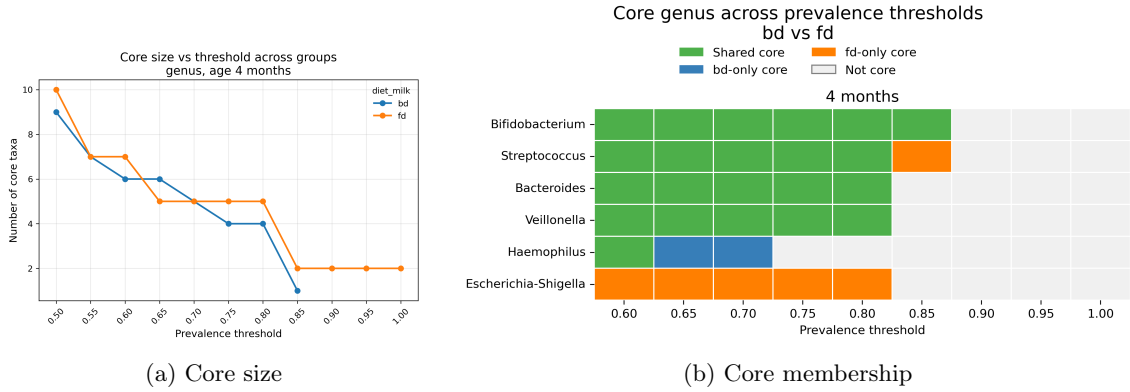


Figure 8: **Core microbiota of breast-fed (bd) and formula-fed (fd) infants at 4 months of age.** Core size (A) and genus-level core membership (B) are shown across increasing prevalence thresholds. Genera are classified as shared between diet groups, specific to one group, or not part of the core.

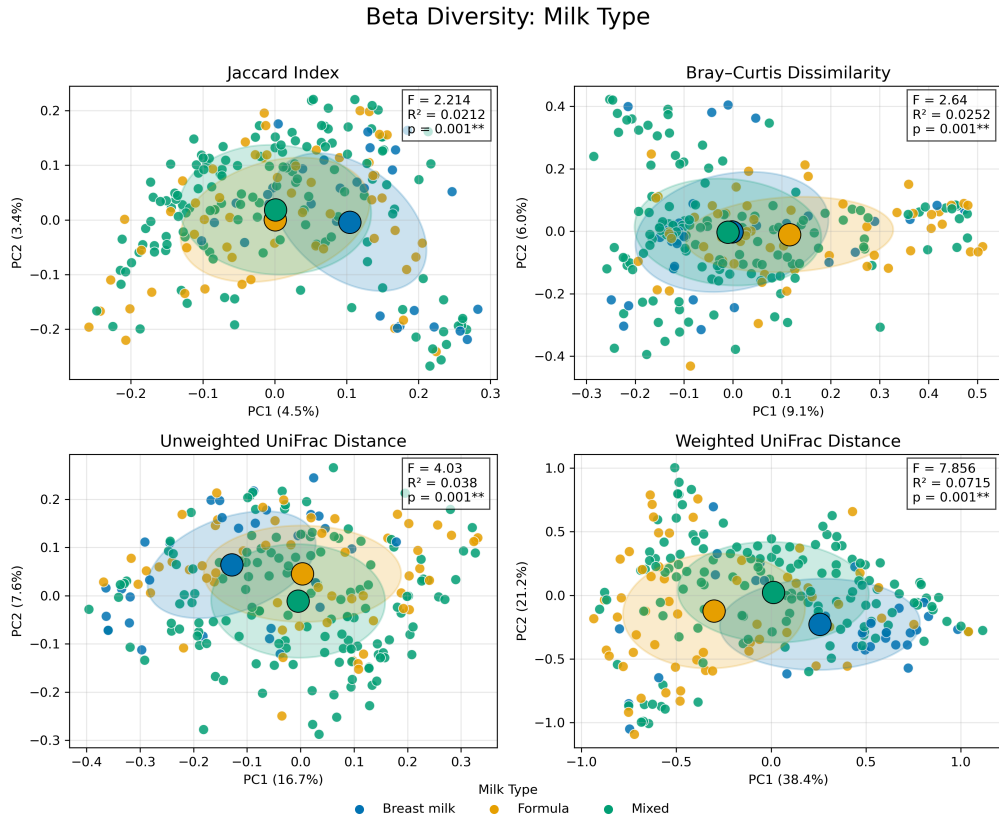


Figure 9: **PCoA plot for beta diversity patterns stratified by feeding type**

Table 3: **Differentially abundant genera associated with milk diet.** Shown are genera identified as differentially abundant by all three methods (DESeq2, MaAsLin2, and ANCOM-BC). Positive log-fold changes indicate enrichment in formula-fed infants compared to breast-fed infants.

Genus	DESeq2 lfc	MaAsLin2 lfc	ANCOM-BC lfc	Direction
<i>Anaerostipes</i>	24.57	3.86	2.11	up
<i>Blautia</i>	20.32	5.20	3.29	up

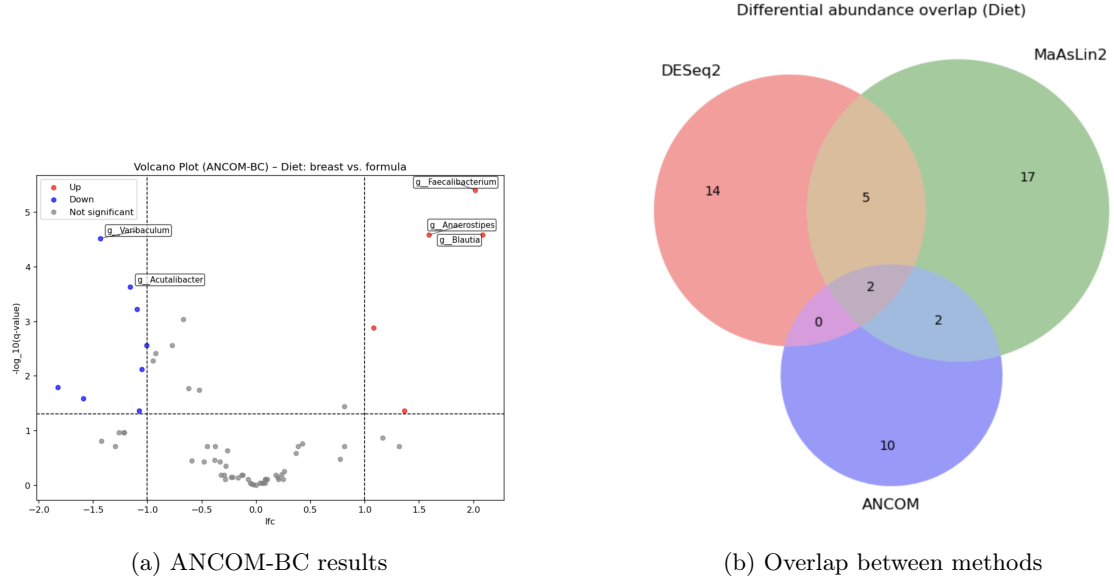


Figure 10: **Differential abundance analysis comparing breast-fed and formula-fed infants.** Volcano plots show genus-level differential abundance results obtained using ANCOM-BC, MaAsLin2, and DESeq2. The Venn diagram summarizes overlap between methods, highlighting taxa consistently identified as differentially abundant across approaches.

4.4 Some of your samples were exposed to some treatment (metadata column "treatment_exposure"). Did the treatment have an effect on the infant gut microbiome? Does the effect of the treatment depend on other covariates?

The treatment exposure is very evenly distributed among our samples (Figure 11). Chi-square tests showed that treatment exposure was strongly associated with geographical location and infant diet, suggesting that these variables are not independent (Supplementary Table 7).

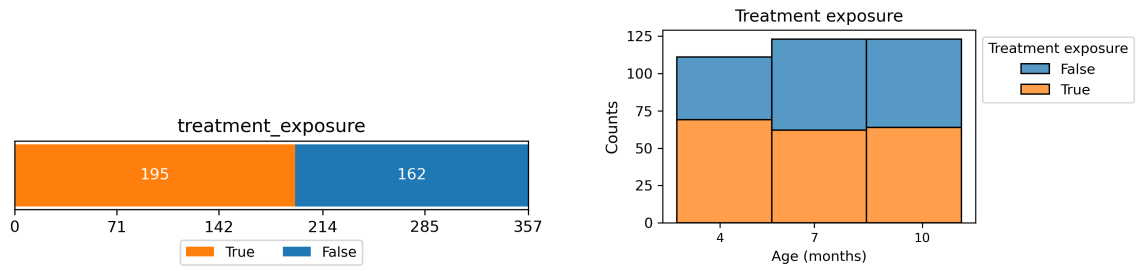
In the differential abundance analysis, *Eggerthella* and *Enterococcus* were depleted in treated infants (Table 4, Supplementary Figure 29). This result should be interpreted with caution, as repeated measurements could not be accounted for in DESeq2, and treatment exposure was strongly associated with geographic location, where both genera were enriched in Russian infants. Both genera include opportunistic pathogens, making their depletion consistent with antibiotic exposure [25, 26].

Treated and untreated infants shared most core genera across ages, indicating limited effects of treatment on core microbiota. However, *Haemophilus* was part of the core microbiota in untreated infants but absent in treated infants (Figure 12), which is consistent with antibiotic treatment targeting pathogens such as *Haemophilus influenzae*, a common cause of respiratory infections in young children [27, 28].

Functional profiling showed that treatment exposure has a very small influence on functional composition, with only three microbial functional pathways significantly depleted (Supplementary Figure 41).

Table 4: **Differentially expressed taxa in treatment exposure.** Down means it's depleted in infants receiving treatment.

Genus	DESeq2 lfc	MaAsLin2 lfc	ANCOM-BC lfc	direction
<i>Eggerthella</i>	-2.53	-1.58	-0.93	down
<i>Enterococcus</i>	-2.85	-1.74	-1.12	down



(a) Overall distribution of treatment exposure across all samples. (b) Distribution of treatment exposure stratified by infant age (4, 7, and 10 months).

Figure 11: **Overview of treatment exposure metadata.** (a) Total counts of samples with and without reported treatment exposure. (b) Age-stratified distribution of treatment exposure status, illustrating how exposure frequencies vary across early infancy.

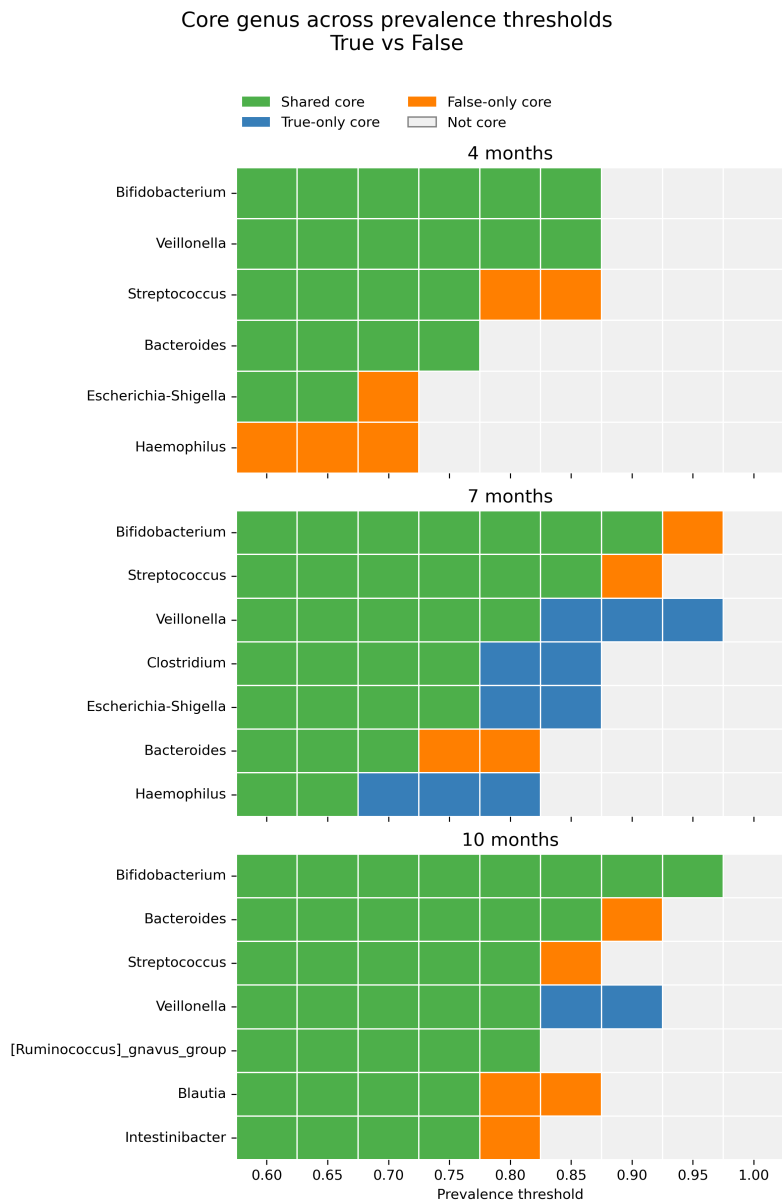


Figure 12: **Core genus membership across prevalence thresholds stratified by treatment exposure and age.** Panels show whether bacterial genera are part of the shared core (present in both treated and untreated groups), treatment-only core, control-only core, or not part of the core, across increasing prevalence thresholds (0.60–1.00) at 4, 7, and 10 months of age.

5 Discussion

In this project, we investigated the development of the infant gut microbiome during the first year of life, focusing on the effects of age, diet, geographic location, treatment exposure, and delivery mode. Overall, we found that age is the primary driver of infant gut microbiome development. This is characterized by increasing diversity and enrichment of anaerobic genera such as *Blautia*, *Roseburia* and *Thomasclavelia*. Diet and geographic location also shape microbial composition, while treatment exposure appears to have more limited and targeted effects.

Primer choice and genus-level taxonomic resolution may have biased detection of certain taxa, and de novo phylogenetic reconstruction introduces additional uncertainty.

Beta diversity analyses revealed significant differences across age groups and dietary categories, particularly for phylogenetic distance metrics. However, these results should be interpreted cautiously, as PERMANOVA primarily tests differences in group centroids and is sensitive to unequal group sizes.

Also, chi-square tests showed that several metadata variables were not independent, most notably geographic location, diet, and treatment exposure, indicating potential confounding in these group-based comparisons. Given the uneven distribution of samples across metadata variables, other statistical tests that allow multiple variables to be assessed jointly could provide more robust results.

Our longitudinal analysis using mixed-effects models identified age-related increases in specific taxa, especially *Blautia*, but the power of these analyses was constrained by sparse longitudinal sampling for many infants and by the use of discrete age categories rather than continuous time. Future analyses could be improved by using first differences and first distances.

Differential abundance analyses identified age- and treatment-associated taxa, but there were several limitations. Not all methods were able to account for repeated measurements, and strong associations between metadata variables such as treatment exposure and geographic location introduced potential confounding. We did account for differences between statistical methods by using a consensus set when looking at taxa, and would use this approach in the functional differential abundance analysis as well in the future.

Core microbiota analyses showed substantial overlap across groups, with group-specific differences mainly apparent at lower prevalence thresholds. Extending this analysis to assess longer-term effects of early diet, such as comparing infants breastfed early in life with those consistently formula-fed at later ages, could provide further insight into changes in the core microbiota during infancy.

Finally, functional profiling using PICRUSt2 gave insights into potential functional consequences of taxonomic differences. However, functional profiles inferred from amplicon sequencing data represent predicted metabolic potential and can substantially differ from those obtained using shotgun metagenomic sequencing [20, 29]. Additionally, prediction accuracy depends on how well the detected ASVs are represented by the genomes in the reference database. Since this dataset contained human gut microbiome data and the NSTI scores were overall low, we suspect that the detected ASVs are well represented in the reference genome database. Functional predictions are also constrained by the functional reference databases used, which capture only a subset of microbial functional diversity.

5.1 Code Availability

All code, as well as additional results and interpretations, are available on our GitHub repository (https://github.com/MasjaHoogendoorn/MicrobiomeAnalysis_TummyTribe)

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6 Appendix

6.1 Metadata

Table 5: **Association between geographic location and metadata variables.** Chi-square tests were used to assess whether geographic location was independent of selected metadata variables.

Variable	p-value	Significance	Interpretation
Delivery mode	7.87×10^{-1}	ns	No evidence of association
Sex	4.32×10^{-1}	ns	No evidence of association
Milk diet	2.20×10^{-15}	***	Strong association
Weaning diet	3.02×10^{-2}	*	Weak association
Treatment exposure	1.01×10^{-11}	***	Strong association
Age (months)	2.18×10^{-4}	***	Strong association

Table 6: **Association between milk diet and metadata variables.** Chi-square tests were used to assess whether milk diet was independent of selected metadata variables.

Variable	p-value	Significance	Interpretation
Delivery mode	7.21×10^{-1}	ns	No evidence of association
Sex	3.84×10^{-1}	ns	No evidence of association
Weaning diet	1.78×10^{-17}	***	Strong association
Geographic location	2.20×10^{-15}	***	Strong association
Treatment exposure	8.93×10^{-4}	***	Strong association
Age (months)	2.50×10^{-15}	***	Strong association

Table 7: **Association between treatment exposure and metadata variables.** Chi-square tests were used to assess whether treatment exposure was independent of selected metadata variables.

Variable	p-value	Significance	Interpretation
Geographic location	1.01×10^{-11}	***	Strong association
Delivery mode	1.96×10^{-1}	ns	No evidence of association
Sex	8.91×10^{-2}	ns	Weak / borderline association
Milk diet	8.93×10^{-4}	***	Strong association
Weaning diet	4.54×10^{-1}	ns	No evidence of association
Age (months)	1.53×10^{-1}	ns	No evidence of association

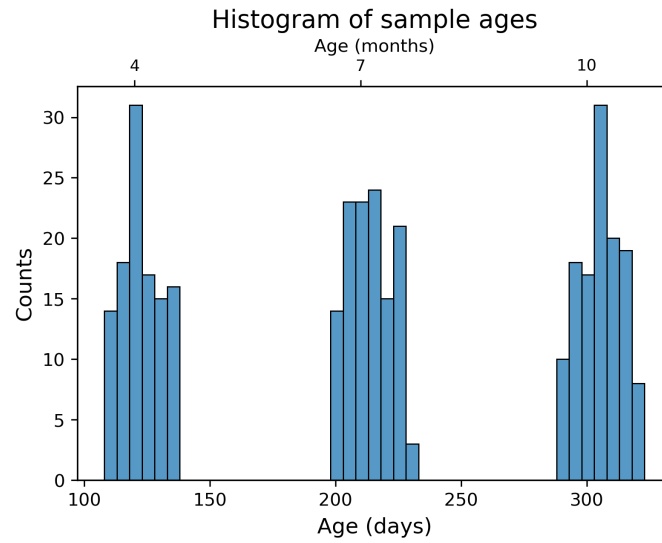
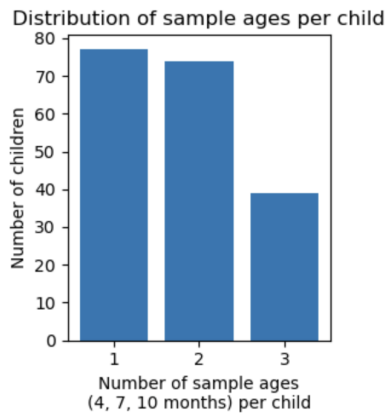
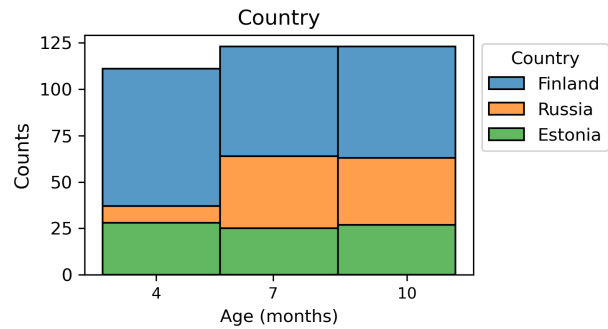


Figure 13: Histogram showing sample distribution of different ages in days and indicating ages in months.



(a) Distribution of sample ages (4, 7, and 10 months) per child



(b) Overall distribution of geographical location categories across all samples.

Figure 14: Overview of infant age and geographical location metadata.

Overview of the distributions in the metadata

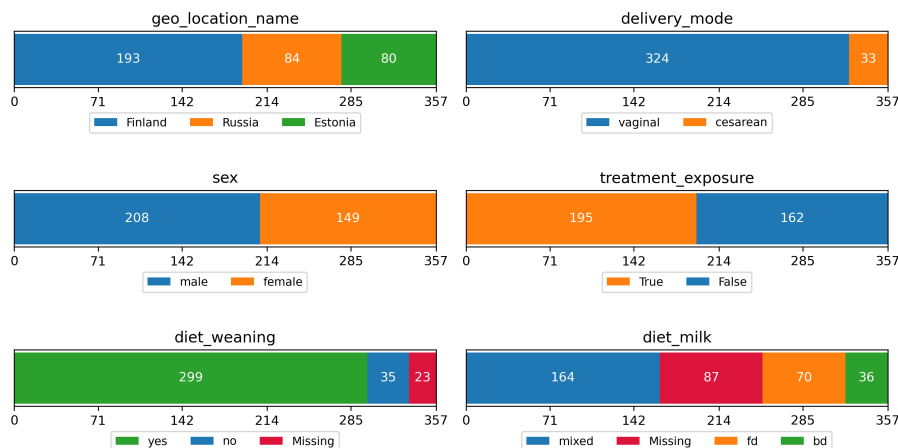


Figure 15: Overview of sample counts for different metadata categories.

Overview of the distributions in the metadata

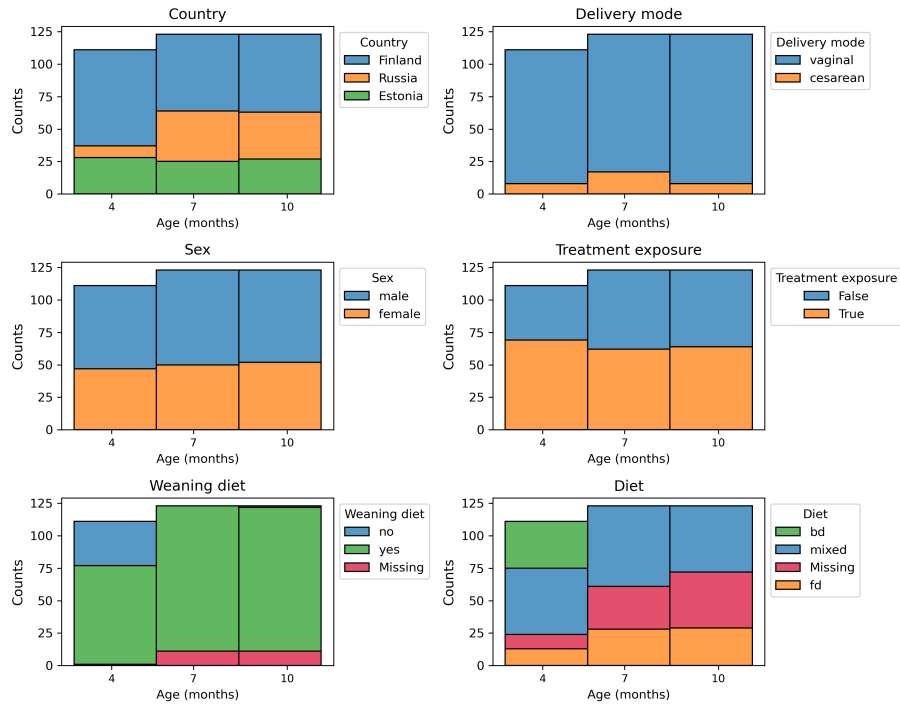


Figure 16: Overview of sample counts for different metadata categories per age group.

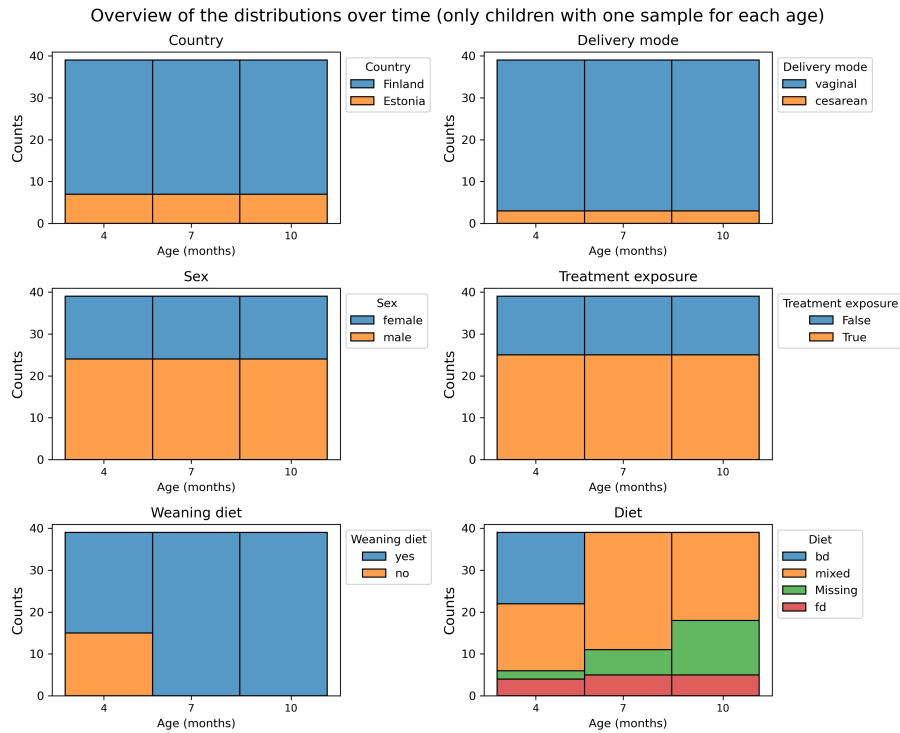


Figure 17: Overview of sample counts for different metadata categories for children that have samples for ages 4, 7 and 10 months.

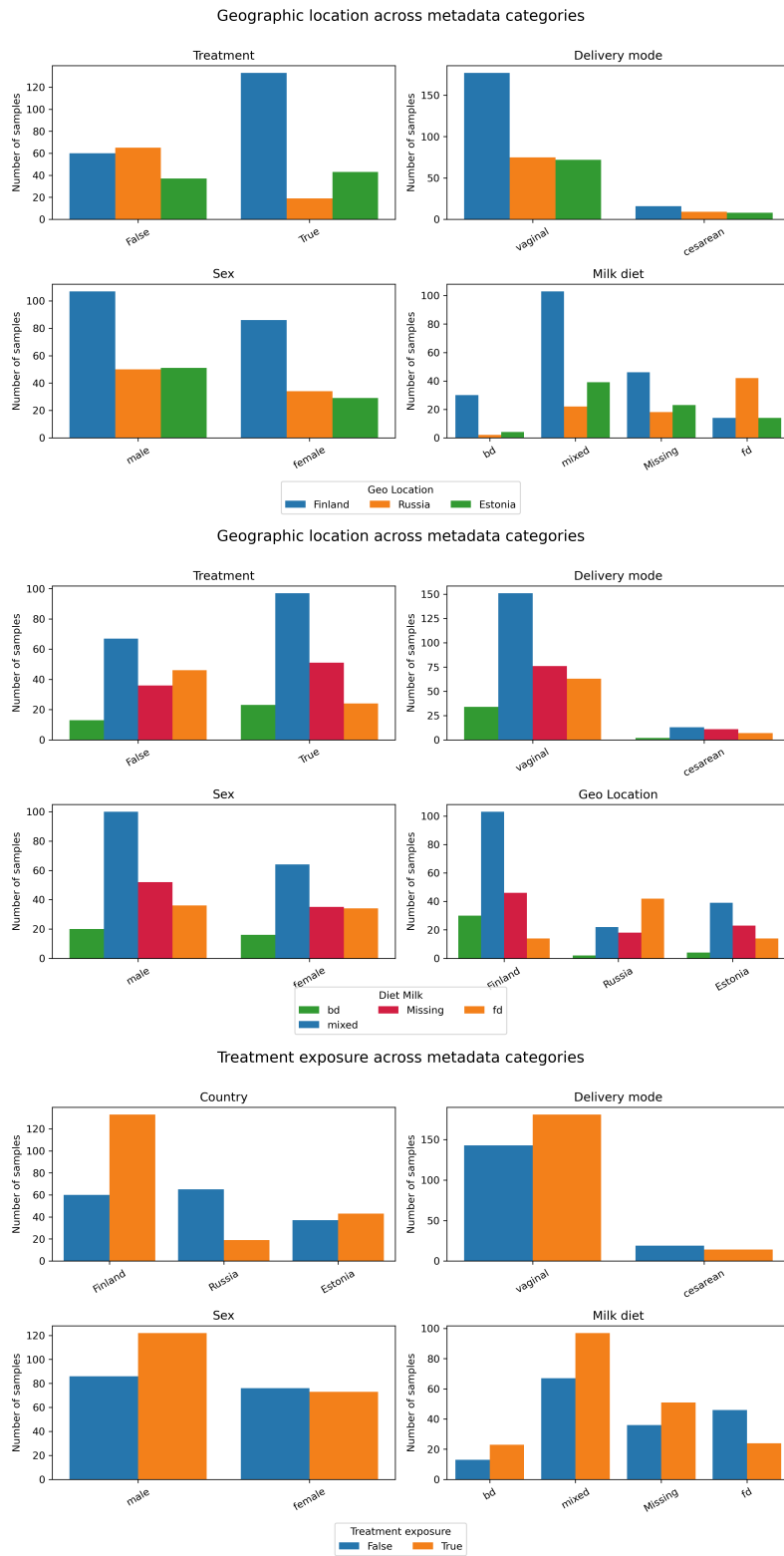


Figure 18: **Distribution of key metadata variables.** Bar plots show sample counts stratified by metadata categories for geographic location, milk diet, and treatment exposure.

6.2 Quality Control

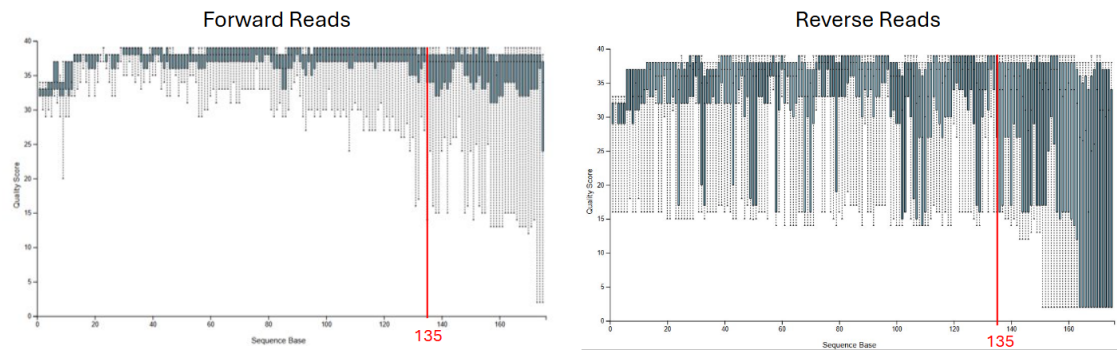
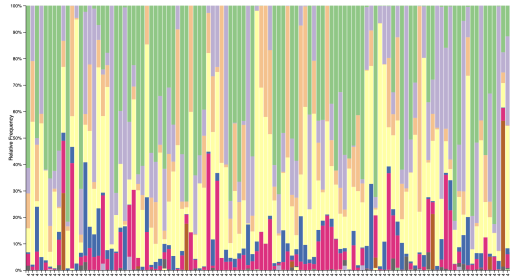
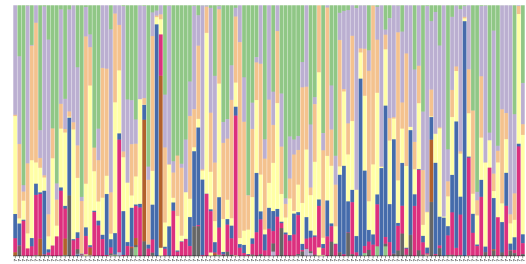


Figure 19: Read Quality and truncation lengths.

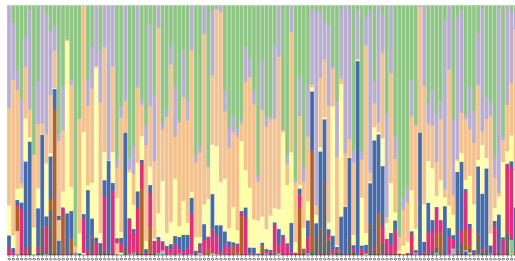
6.3 Taxonomy



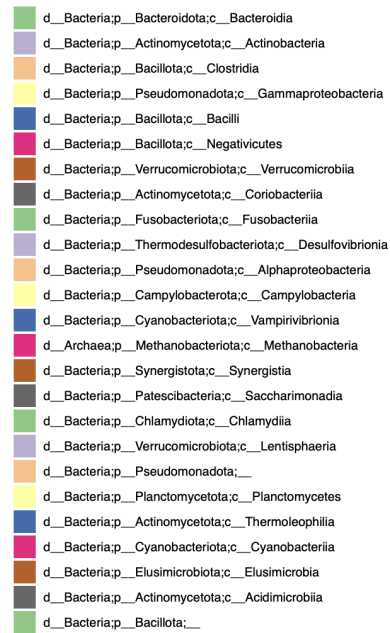
(a) Taxonomy level 3 bar plot for samples of age 4 months.



(b) Taxonomy level 3 bar plot for samples of age 7 months.



(c) Legend for taxonomy bar plots.



(d)

Figure 20: Taxonomy level 3 (class) bar plots for ages 4, 7, and 10 months including legend. Taxonomy bar plots for all other taxonomic levels can be viewed on our GitHub in notebook '03_taxonomy'.

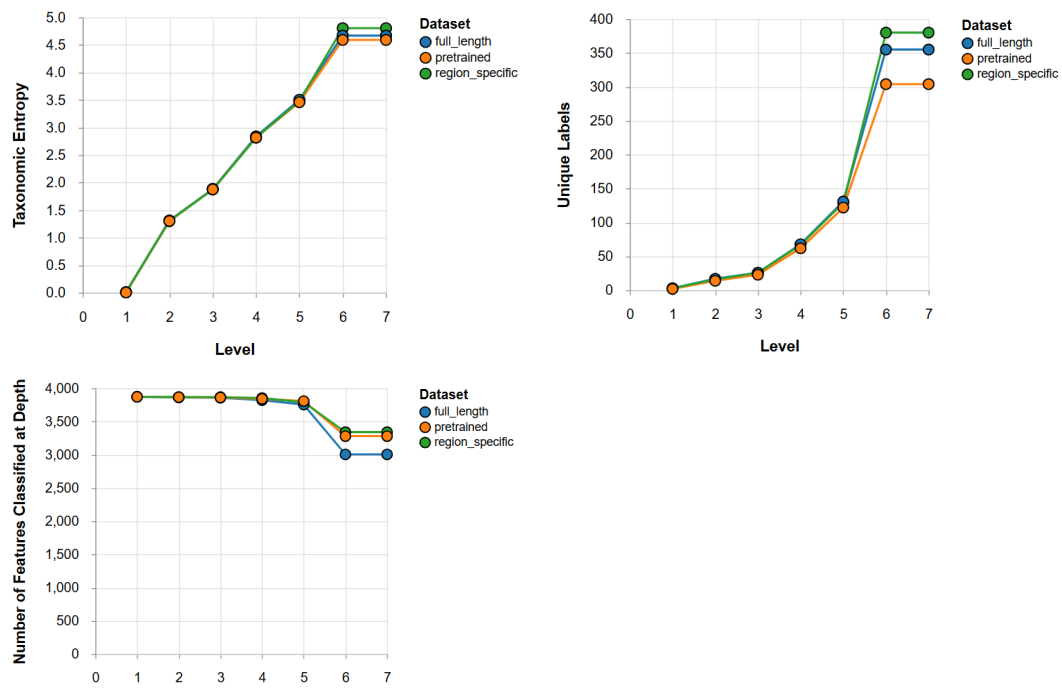
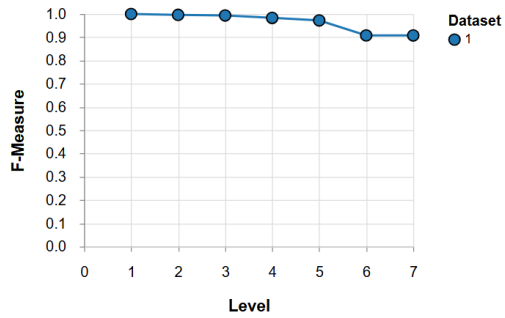
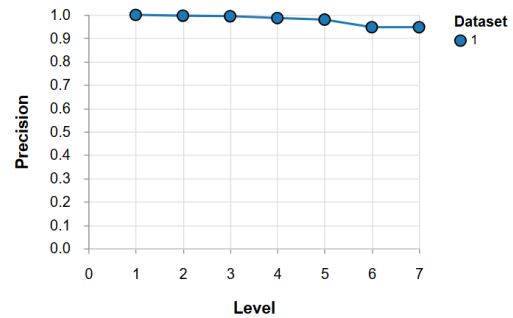


Figure 21: Comparison between full-length, pretrained, and region-specific classifiers.

REScriPt Evaluate Classifications



REScriPt Evaluate Classifications



REScriPt Evaluate Classifications

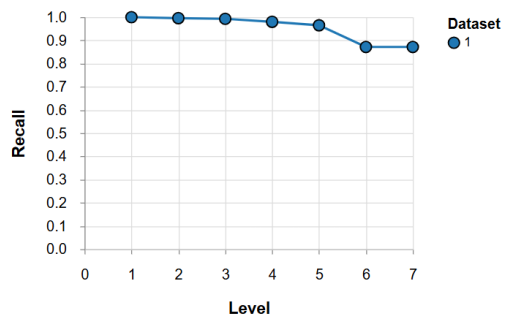
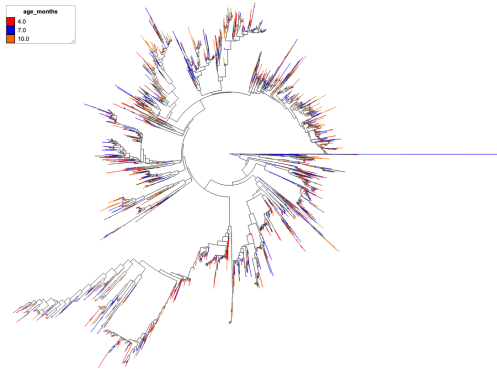
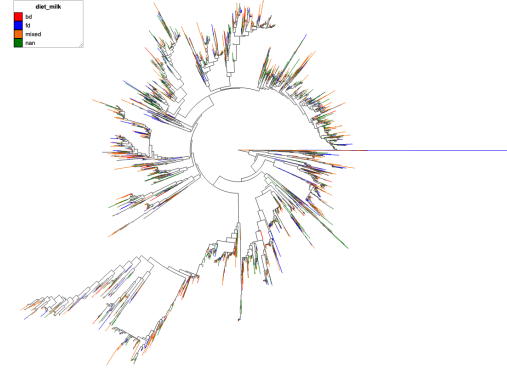


Figure 22: Evaluation of V4 region-specific classifier.

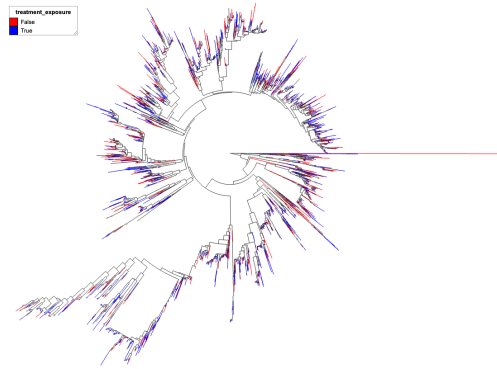
6.4 Phylogeny



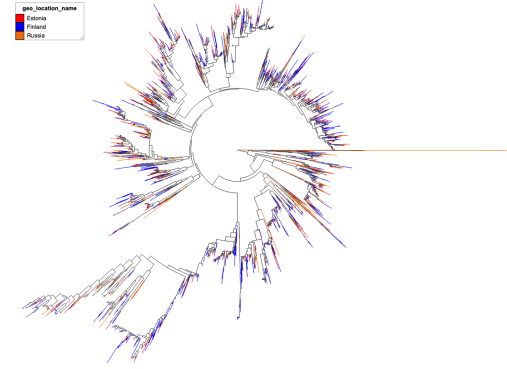
(a) Phylogenetic de novo tree coloured by age.



(b) Phylogenetic de novo tree coloured by diet.

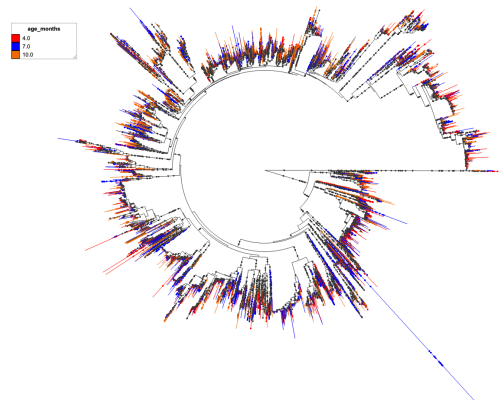


(c) Phylogenetic de novo tree coloured by treatment exposure.

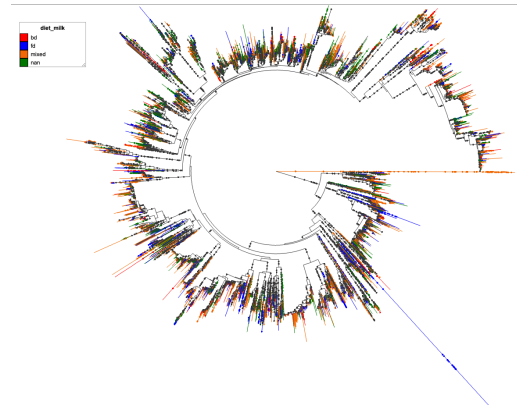


(d) Phylogenetic de novo tree coloured by geographical location.

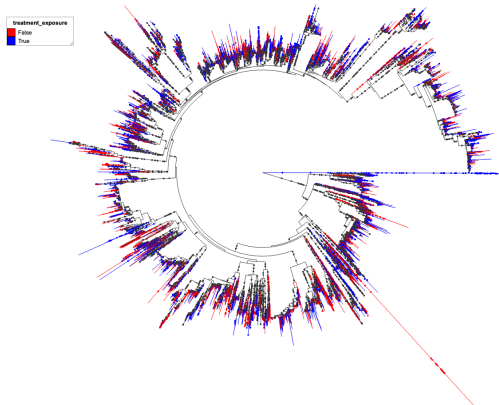
Figure 23: Phylogenetic de novo trees coloured by age, diet, treatment exposure and geographical location.



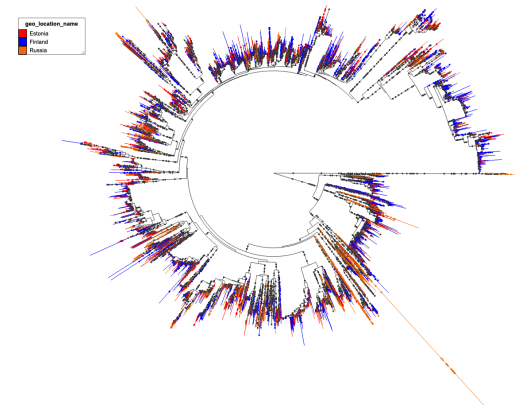
(a) Phylogenetic Fragment Insertion tree coloured by age.



(b) Phylogenetic Fragment Insertion tree coloured by diet.



(c) Phylogenetic Fragment Insertion tree coloured by treatment exposure.



(d) Phylogenetic Fragment Insertion tree coloured by geographical location.

Figure 24: Phylogenetic Fragment Insertion trees coloured by age, diet, treatment exposure and geographical location.

6.5 Diversity Analysis

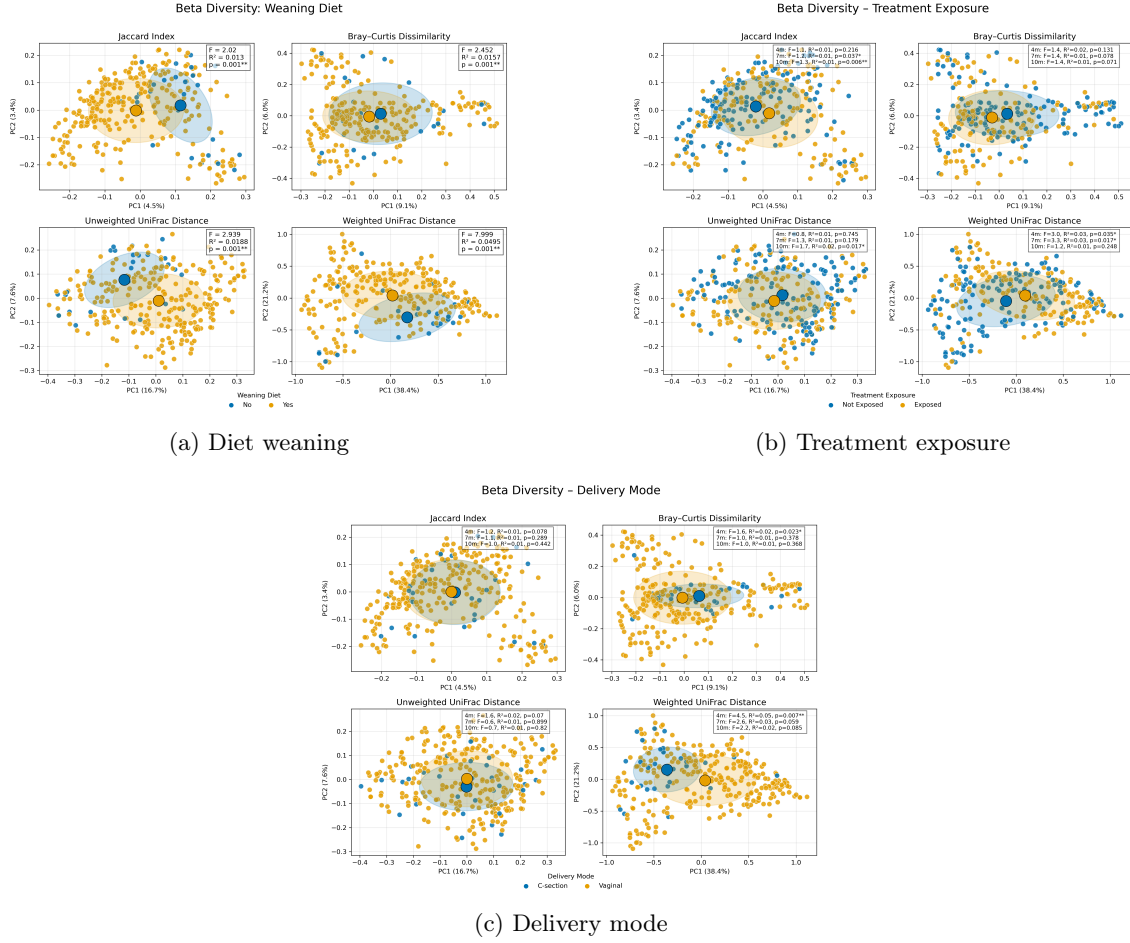


Figure 25: **Beta diversity stratified by diet weaning (time-variant), treatment exposure (time-invariant) and delivery mode (time-invariant).** For time-variant variables, group differences were tested using PERMANOVA (adonis2) with infant ID included as a strata variable to account for repeated measurements. For time-invariant variables, PERMANOVA (adonis2) was performed separately for each age group using one sample per infant.

6.6 Longitudinal Analysis

Table 8: **Convergence of linear mixed-effects models for alpha diversity metrics.** Linear mixed-effects (LME) models were fitted for each alpha diversity metric and predictor. Model convergence is indicated for each combination.

Predictor	Faith's PD	Shannon	Evenness
Diet (weaning)	No	Yes	No
Diet (milk)	No	No	No
Geographic location	No	Yes	No
Delivery mode	No	No	Yes
Sex	Yes	Yes	No
Treatment exposure	No	Yes	Yes

Table 9: **Summary of linear mixed-effects models for alpha diversity.** Shown are the presence of an age effect and age-by-predictor interaction, together with corresponding p-values, for models that successfully converged. For geographic location, Finland and Russia are shown as separate contrasts relative to the reference level (Estonia).

Metric	Predictor	Age effect	Age (p)	Interaction	Age $\times$ Predictor (p)
Faith's PD	Sex	Yes	$< 1 \times 10^{-3}$	Yes	1.0×10^{-3}
Shannon	Diet weaning	Yes	3.4×10^{-2}	No	2.3×10^{-1}
Shannon	Geo location (Finland vs Estonia)	Yes	$< 1 \times 10^{-3}$	Yes	1.0×10^{-2}
Shannon	Geo location (Russia vs Estonia)	Yes	$< 1 \times 10^{-3}$	Yes	9.0×10^{-3}
Shannon	Sex	Yes	$< 1 \times 10^{-3}$	No	6.1×10^{-1}
Shannon	Treatment exposure	Yes	$< 1 \times 10^{-3}$	No	9.0×10^{-1}
Evenness	Delivery mode	No	8.2×10^{-1}	No	3.4×10^{-1}
Evenness	Treatment exposure	No	7.3×10^{-2}	No	8.8×10^{-1}

Table 10: **Most important taxa identified by the classification model.** Taxa are reported at the genus level when available, together with their corresponding taxonomic class and relative importance scores.

Class	Genus	Importance
Bacilli	<i>Thomasclavelia</i>	0.1520
Clostridia	<i>Blautia</i>	0.1051
Gammaproteobacteria	<i>Escherichia-Shigella</i>	0.0253
Bacilli	<i>Lacticaseibacillus</i>	0.0242
Clostridia	Unclassified (Lachnospiraceae)	0.0240
Clostridia	<i>Flavonifractor</i>	0.0238
Clostridia	<i>Faecalibacterium</i>	0.0213

Table 11: **Convergence of linear mixed-effects models for selected taxa.** Linear mixed-effects models were fitted for each taxon and predictor. Model convergence is indicated for each combination.

Predictor	Thomasclavelia	Blautia
Diet (weaning)	Yes	No
Diet (milk)	No	No
Geographic location	Yes	Yes
Delivery mode	Yes	Yes
Sex	No	Yes
Treatment exposure	No	Yes

Table 12: **Linear mixed-effects model results for selected taxa.** Shown are the presence of an age effect and age-by-predictor interaction, together with corresponding p-values, for converged models only. For geographic location, Finland and Russia are shown as separate rows and interpreted relative to the reference level (Estonia).

Taxon	Predictor	Age effect	Age (p)	Interaction	Age $\times$ Predictor (p)
Thomasclavelia	Geo location (Finland vs Estonia)	Yes	6.0×10^{-3}	Yes	1.3×10^{-2}
Thomasclavelia	Geo location (Russia vs Estonia)	Yes	6.0×10^{-3}	Yes	2.2×10^{-2}
Thomasclavelia	Delivery mode	No	9.6×10^{-1}	No	7.7×10^{-1}
Blautia	Geo location (Finland vs Estonia)	Yes	3.8×10^{-2}	No	5.9×10^{-1}
Blautia	Geo location (Russia vs Estonia)	Yes	3.8×10^{-2}	No	2.4×10^{-1}
Blautia	Delivery mode	No	5.3×10^{-2}	No	4.4×10^{-1}
Blautia	Sex	Yes	3.5×10^{-2}	No	3.6×10^{-1}
Blautia	Treatment exposure	Yes	1.1×10^{-2}	No	6.0×10^{-1}

6.7 Differential Abundance Analysis

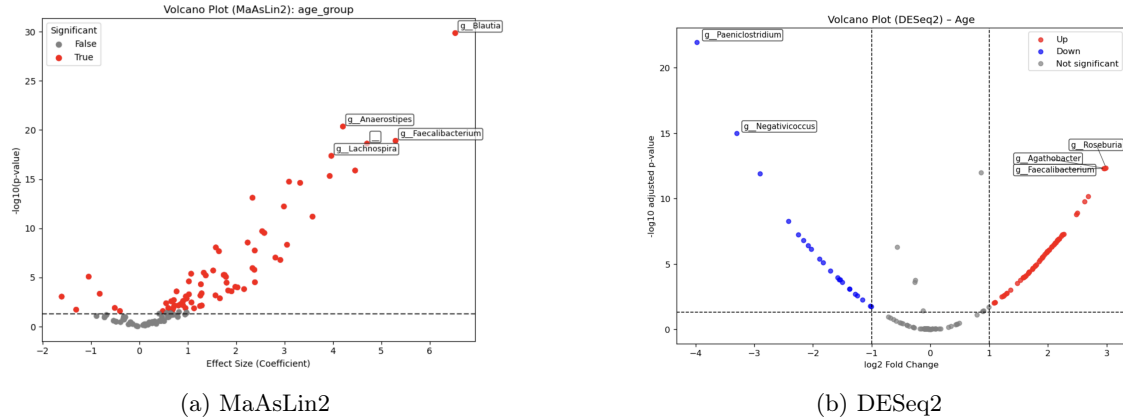
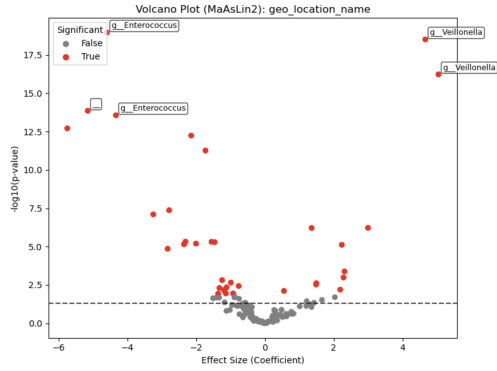
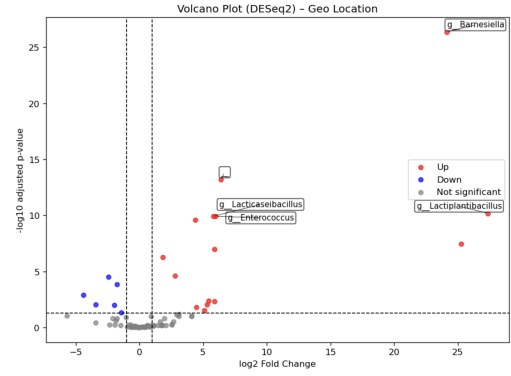


Figure 26: **Volcano Plots from MaAsLin2 and DESeq2 comparing age groups 4 months and 10 months.**

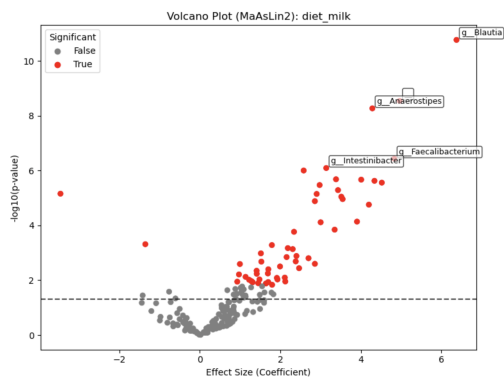


(a) MaAsLin2

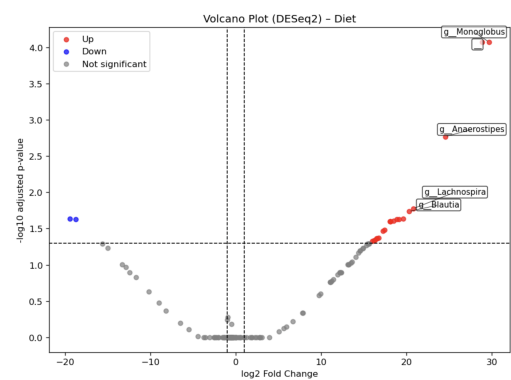


(b) DESeq2

Figure 27: Volcano Plots from MaAsLin2 and DESeq2 comparing geographical locations Finland and Russia.

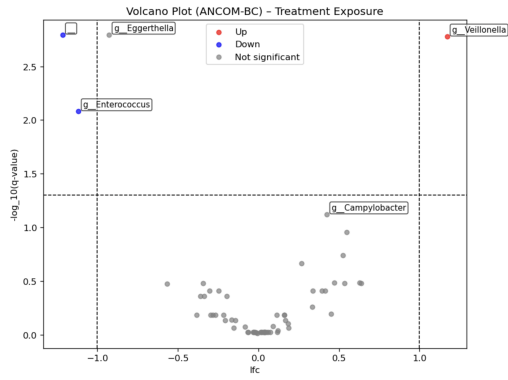


(a) MaAsLin2

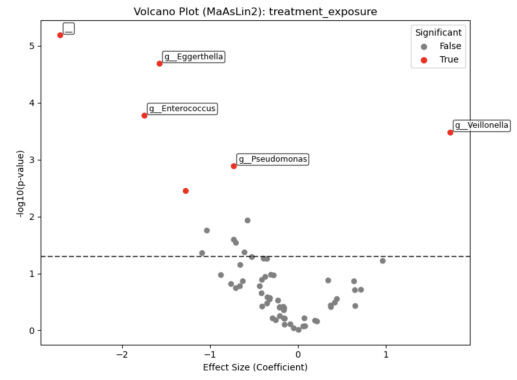


(b) DESeq2

Figure 28: Volcano Plots from MaAsLin2 and DESeq2 comparing breast-fed and formula-fed groups.

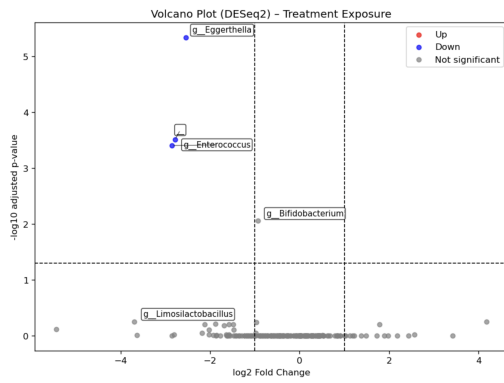


(a) ANCOM-BC

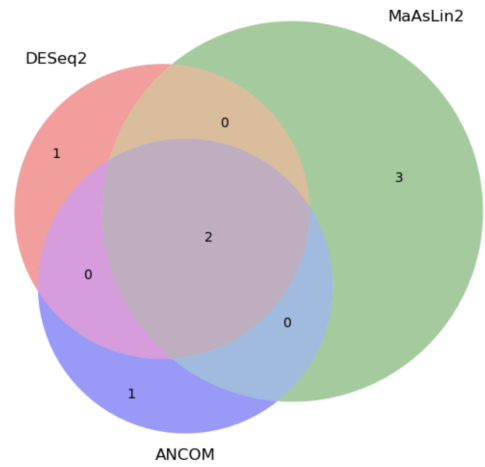


(b) MaAsLin2

Differential abundance overlap (Treatment)

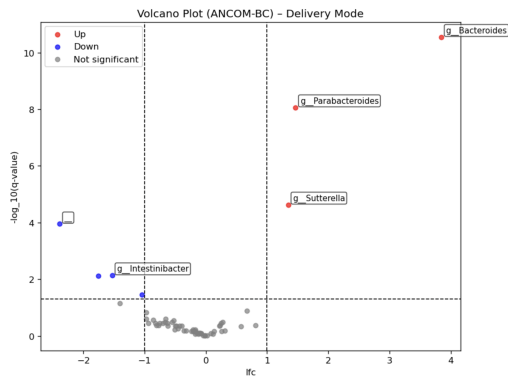


(c) DESeq2

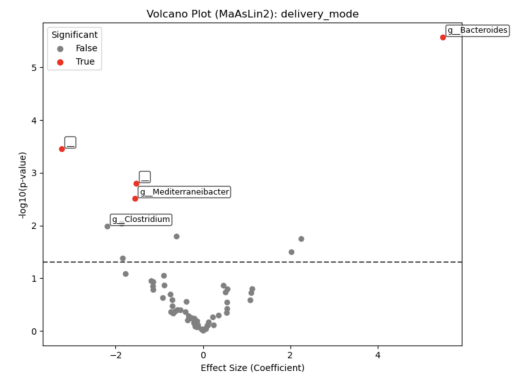


(d) Venn Diagram of consensus set

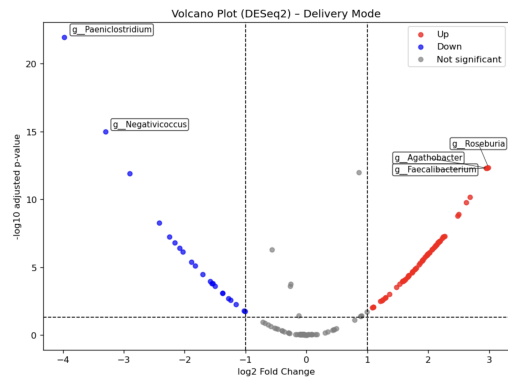
Figure 29: Volcano Plots from ANCOM-BC, MaAsLin2 and DESeq2 and Venn Diagram of consensus set of differentially expressed taxa in treatment exposure. 2 differentially expressed taxa were found by all three methods.



(a) ANCOM-BC



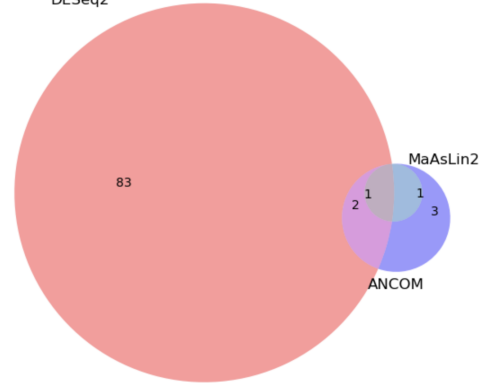
(b) MaAsLin2



(c) DESeq2

Differential abundance overlap (Deliverymode)

DESeq2



(d) Venn Diagram of consensus set

Figure 30: Volcano Plots from ANCOM-BC, MaAsLin2 and DESeq2 and Venn Diagram of consensus set of differentially expressed taxa in delivery mode. 1 differentially expressed taxa were found by all three methods.

6.8 Core Microbiota Analysis

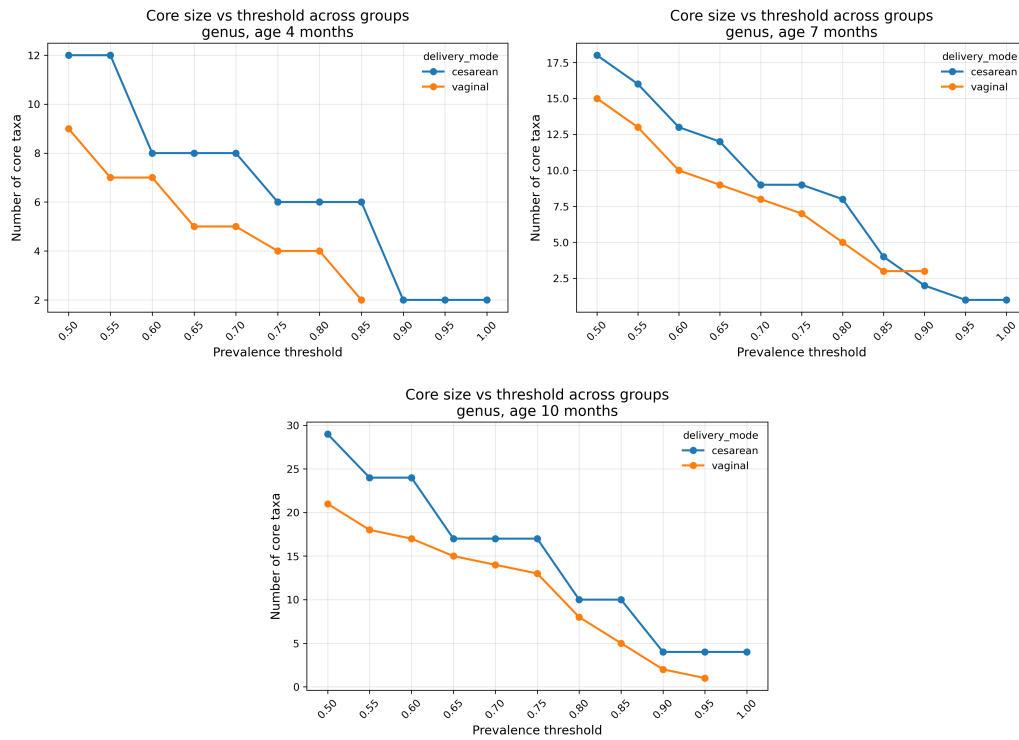


Figure 31: **Core genus size across prevalence thresholds by delivery mode.** Number of core genera as a function of prevalence threshold for cesarean and vaginal delivery, shown separately for ages 4, 7, and 10 months.

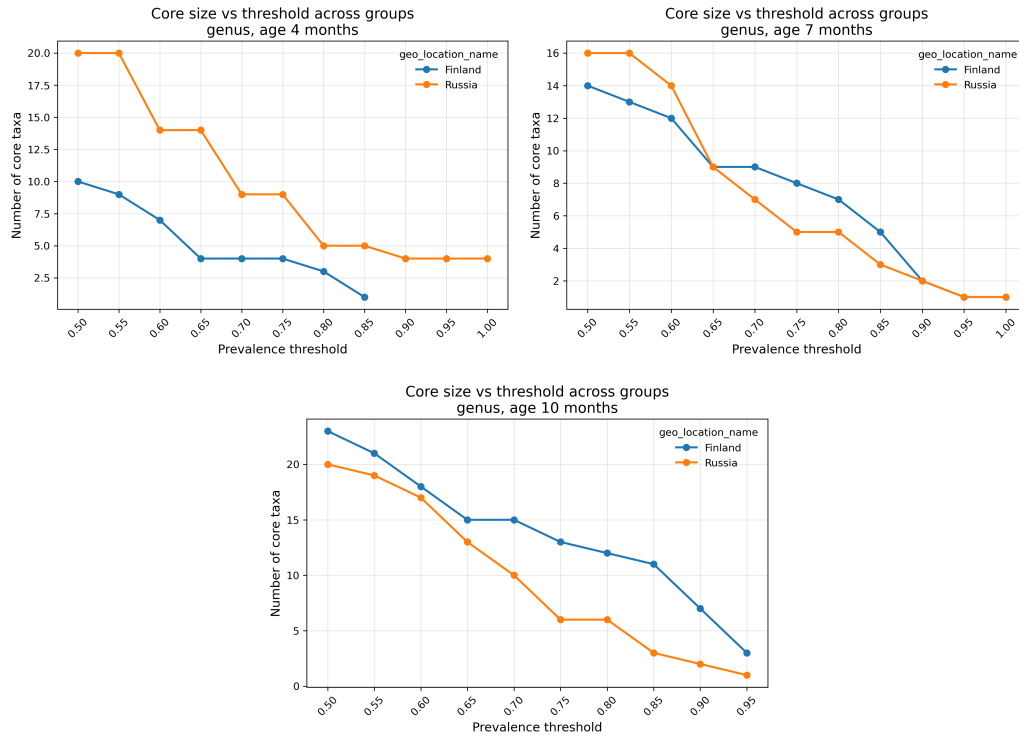


Figure 32: **Core genus size across prevalence thresholds by geographic location.** Core size dynamics across increasing prevalence thresholds for Finland and Russia, shown separately for ages 4, 7, and 10 months.

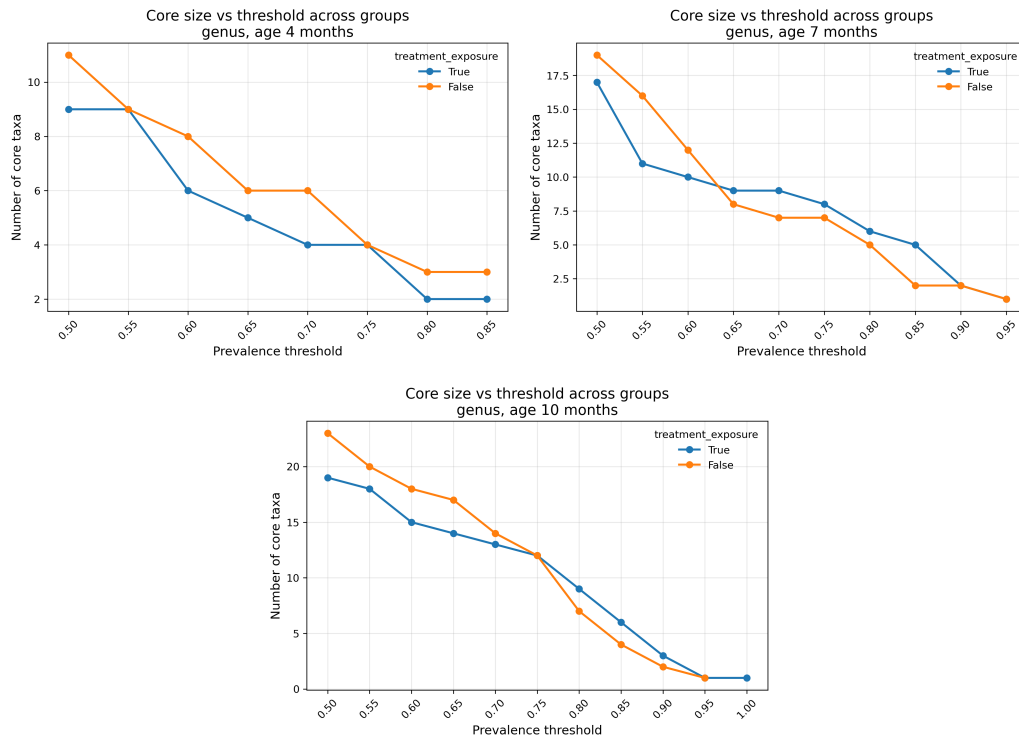


Figure 33: **Core genus size across prevalence thresholds by treatment exposure.** Number of core genera as a function of prevalence threshold for treated and untreated infants, shown separately for ages 4, 7, and 10 months.

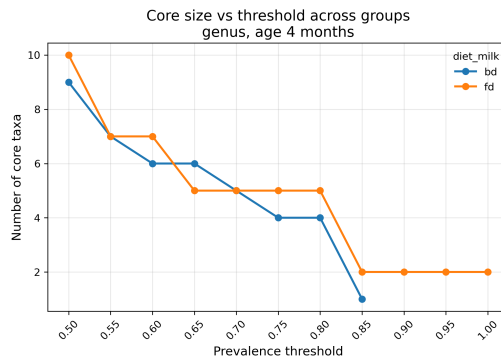


Figure 34: **Core genus size across prevalence thresholds by milk diet.** Core size comparison between breastfed and formula-fed infants at 4 months of age.

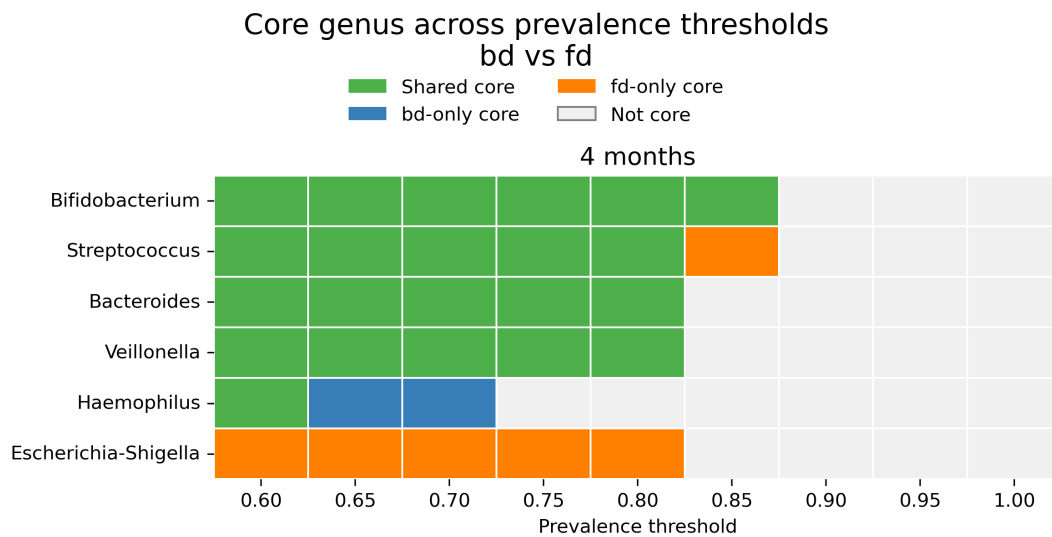


Figure 35: **Core genera across prevalence thresholds by milk diet.** Core genus membership at different prevalence thresholds comparing breastfed (bd) and formula-fed (fd) infants. Genera are classified as shared between groups, specific to one group, or not part of the core. Only one timepoint is shown as infants measured at 7 and 10 months were not breastfed anymore.

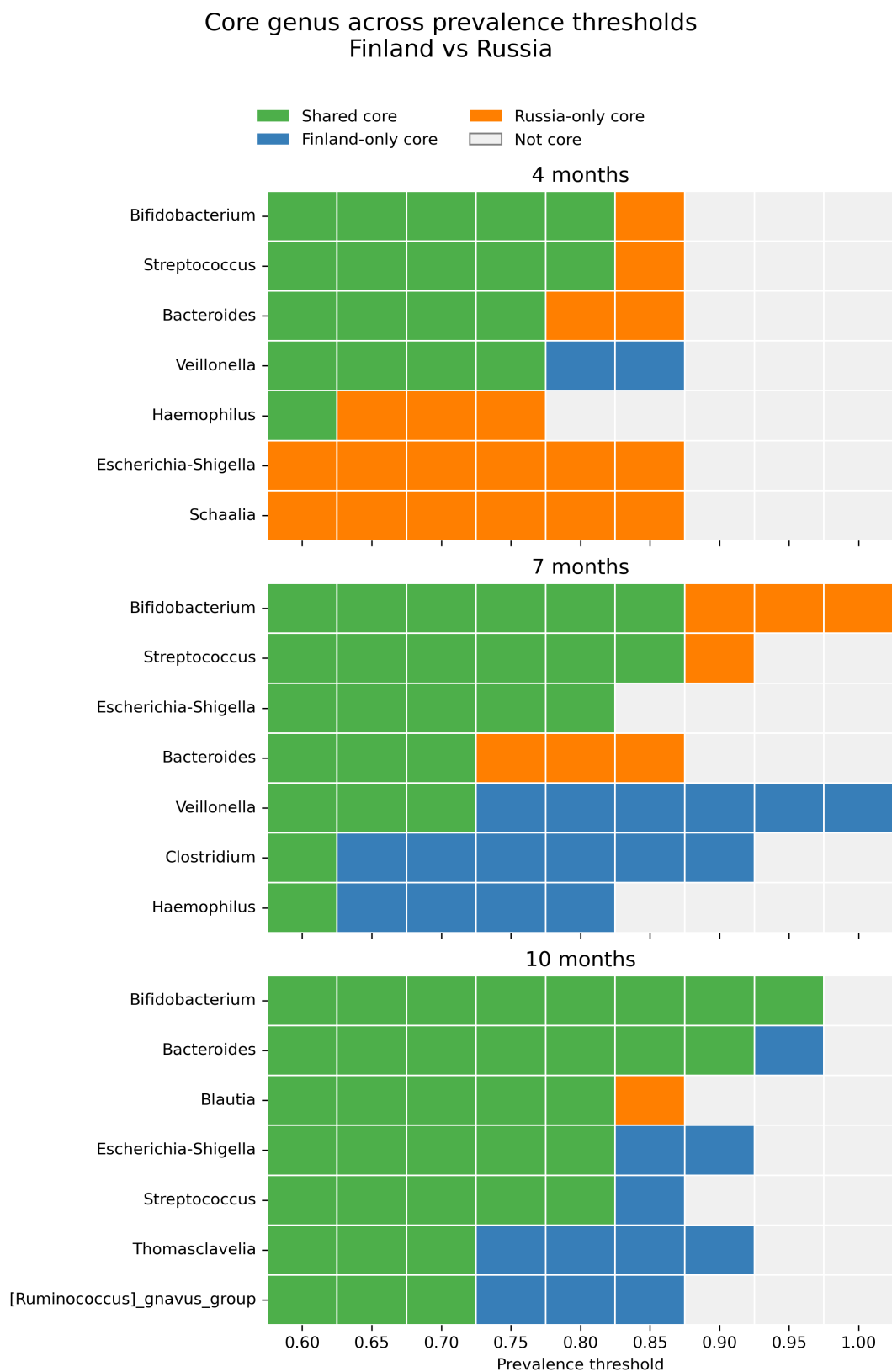


Figure 36: **Core genera across prevalence thresholds by geographic location.** Comparison of core genus membership between Finnish and Russian infants across increasing prevalence thresholds.

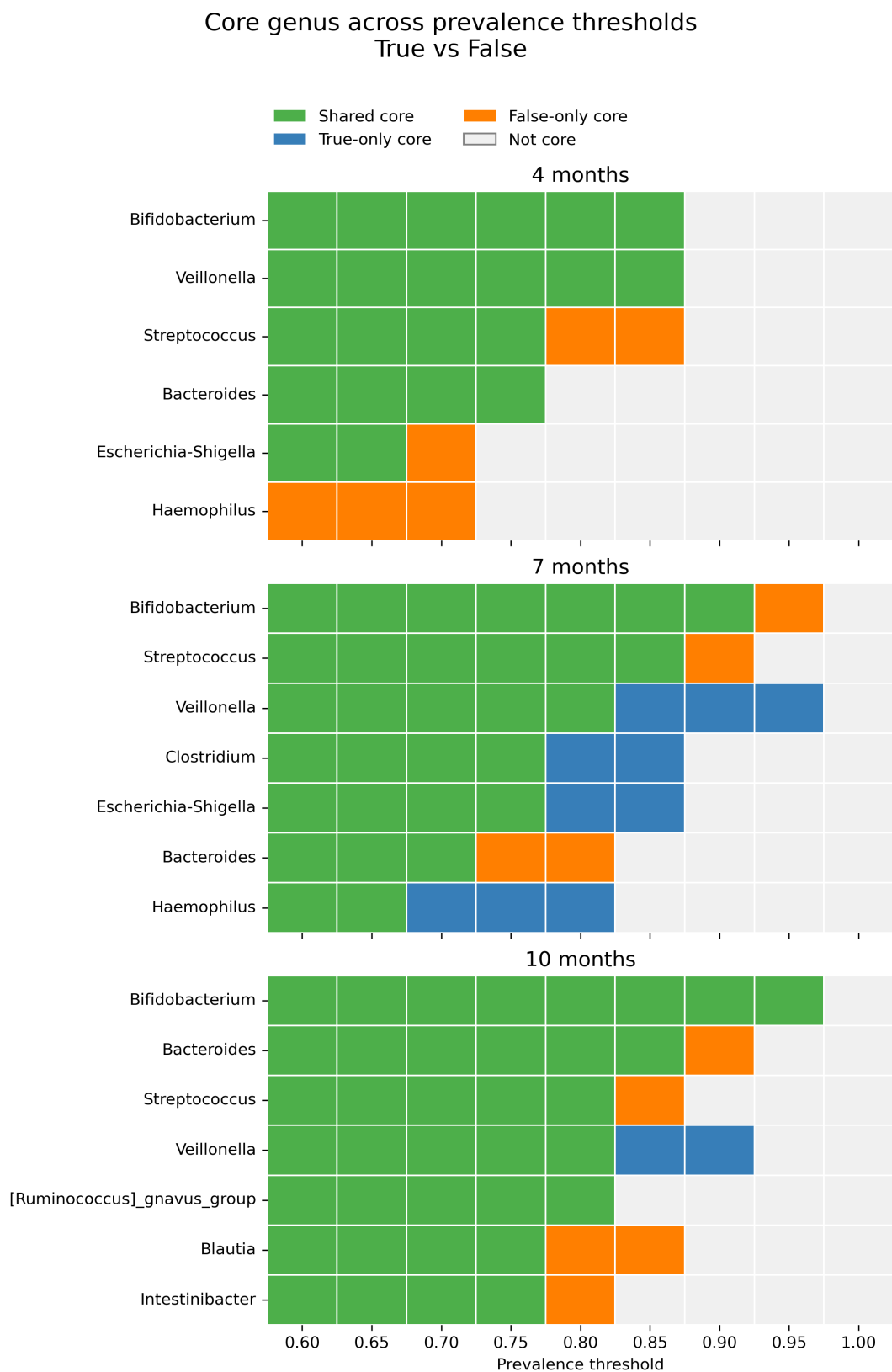


Figure 37: **Core genera across prevalence thresholds by treatment exposure.** Core genus patterns comparing treated and untreated infants across prevalence thresholds, highlighting shared and treatment-specific core taxa.



Figure 38: **Core genera across prevalence thresholds by delivery mode.** Core genus composition across prevalence thresholds comparing infants born by cesarean section and vaginal delivery. Colors indicate shared and group-specific core genera.

6.9 Functional Profiling

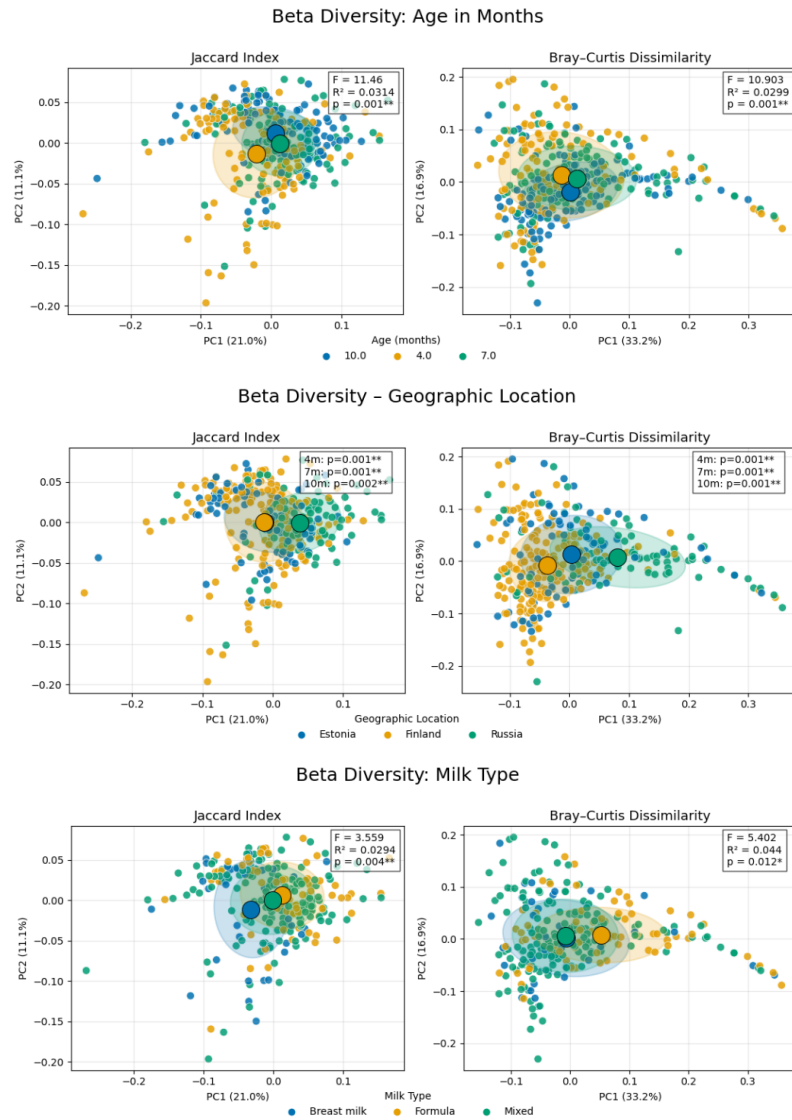


Figure 39: Stacked PCoA plots of functional beta diversity stratified by age group, geographic location and milk type.

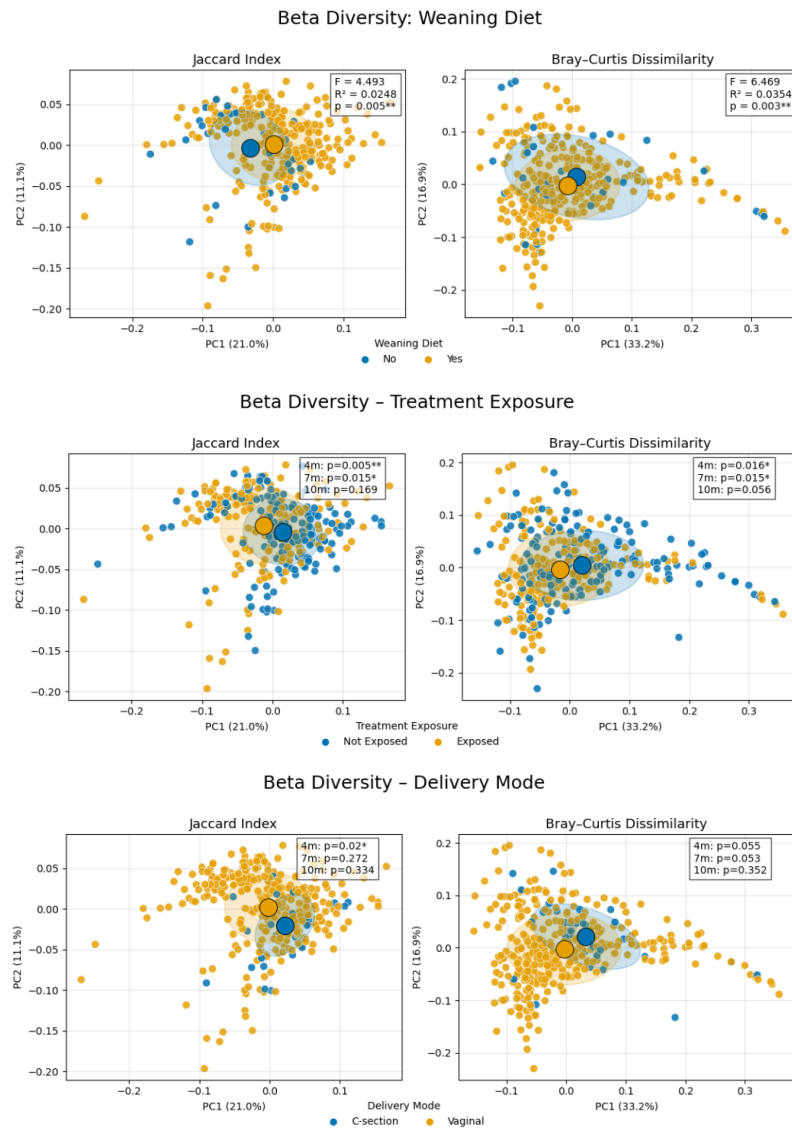
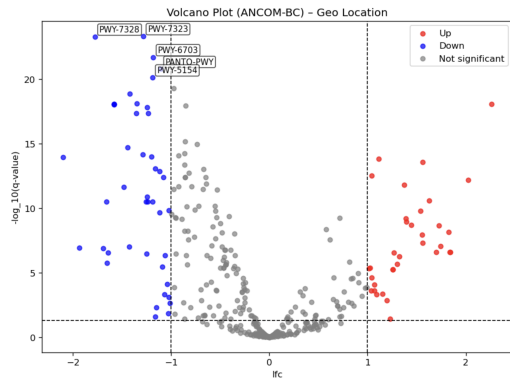
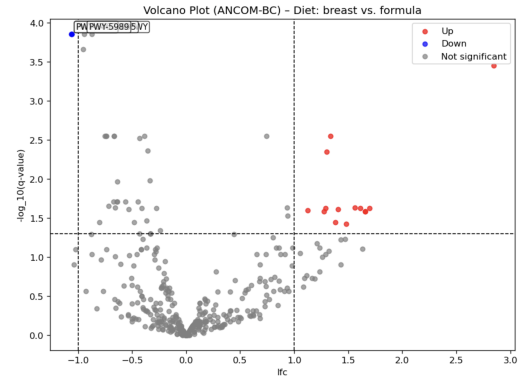


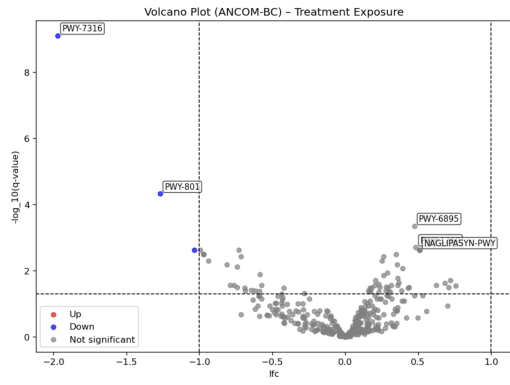
Figure 40: Stacked PCoA plots of functional beta diversity stratified by weaning, treatment exposure and delivery mode



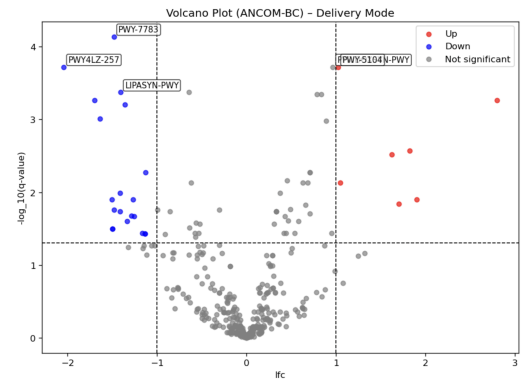
(a) Geographical Location



(b) Diet Milk



(c) Treatment Exposure



(d) Delivery Mode

Figure 41: ANCOM-BC Volcano Plots from Functional Differential Abundance Analysis for changes in geographical location, diet, treatment exposure and delivery mode.